



Chapter 23

Process-to-Process Delivery: UDP, TCP, and SCTP

23-1 PROCESS-TO-PROCESS DELIVERY

The transport layer is responsible for process-to-process delivery—the delivery of a packet, part of a message, from one process to another. Two processes communicate in a client/server relationship, as we will see later.

Topics discussed in this section:

Client/Server Paradigm

Multiplexing and Demultiplexing

Connectionless Versus Connection-Oriented Service

Reliable Versus Unreliable

Three Protocols



The transport layer is responsible for process-to-process delivery.

Figure 23.1 *Types of data deliveries*

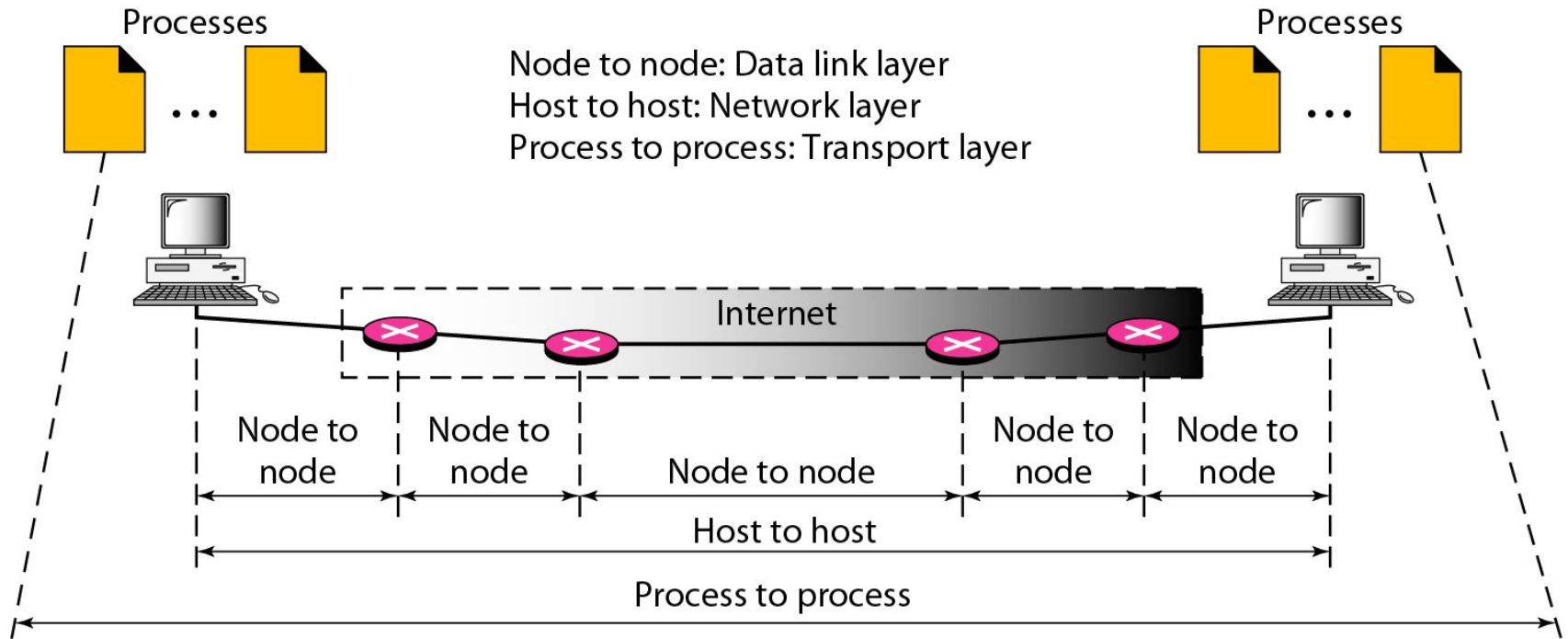


Figure 23.2 *Port numbers*

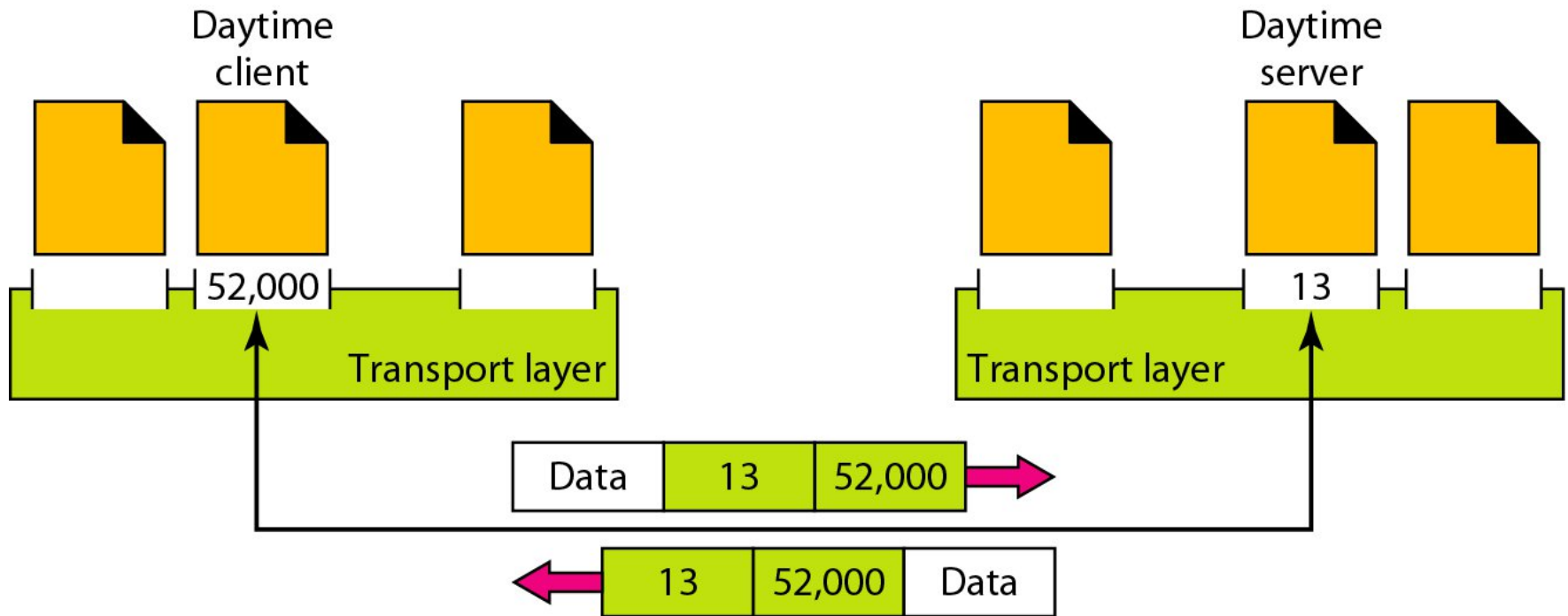


Figure 23.3 *IP addresses versus port numbers*

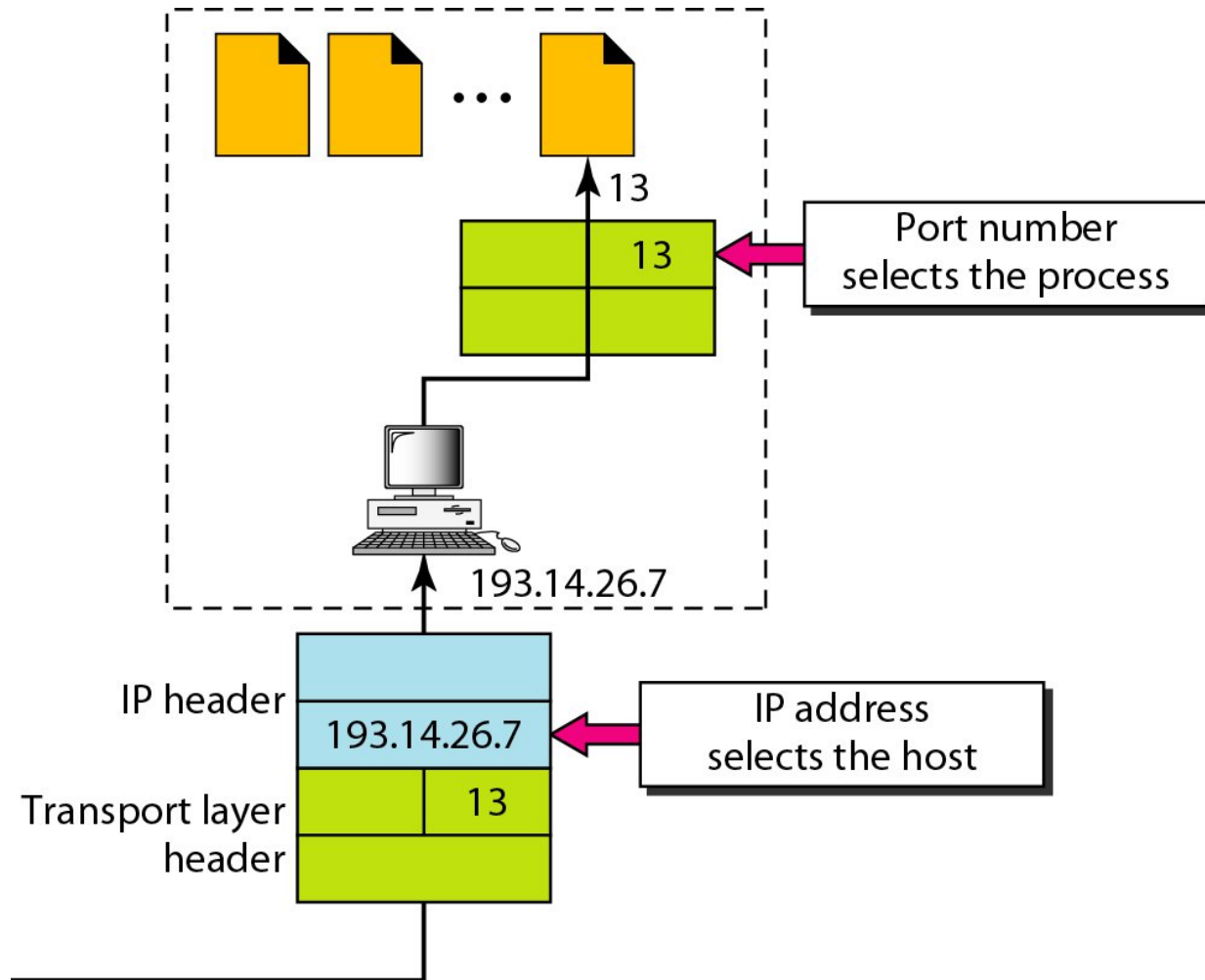


Figure 23.4 *IANA ranges*

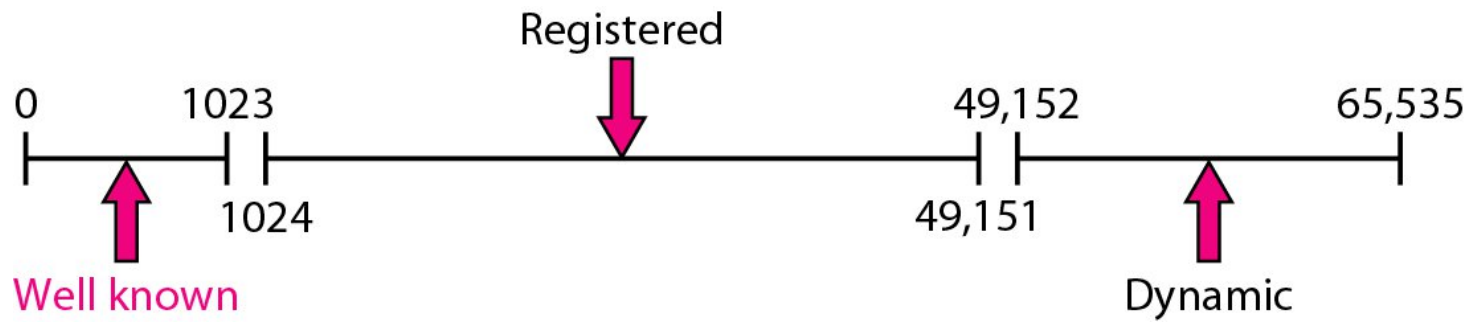


Figure 23.5 *Socket address*

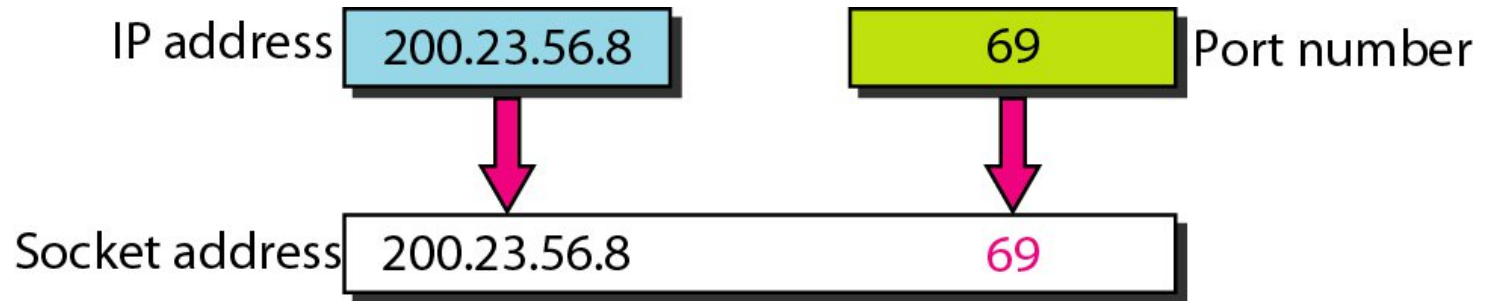


Figure 23.6 *Multiplexing and demultiplexing*

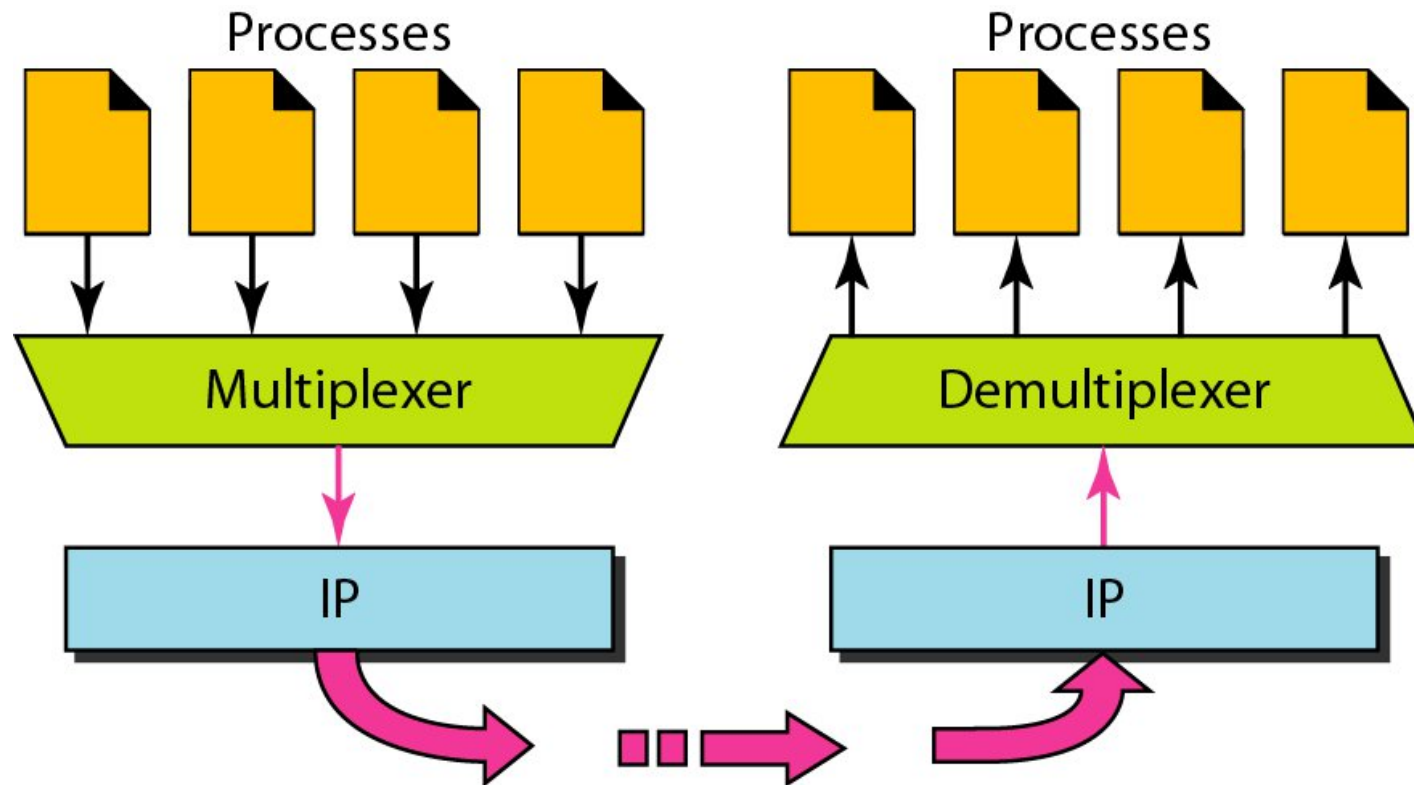


Figure 23.7 *Error control*

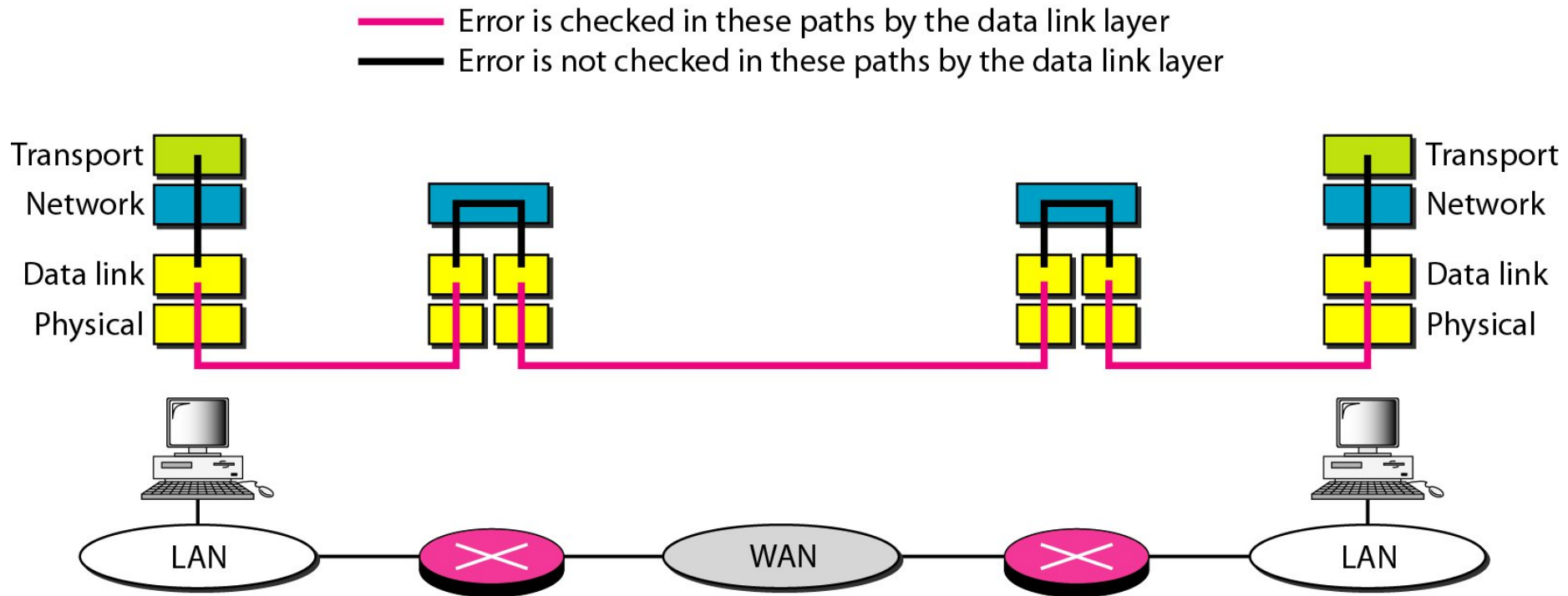
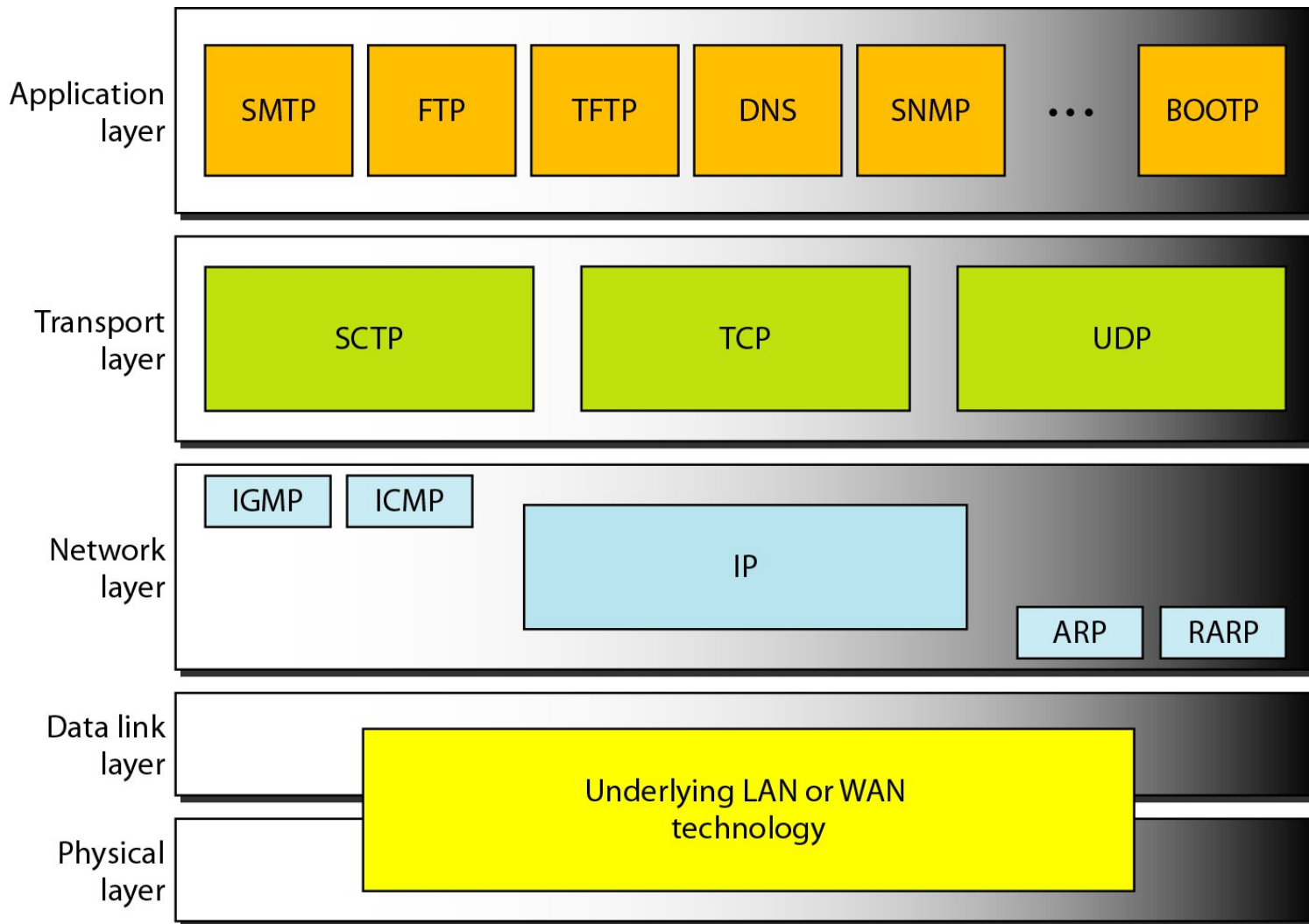


Figure 23.8 *Position of UDP, TCP, and SCTP in TCP/IP suite*



23-2 USER DATAGRAM PROTOCOL (UDP)

The User Datagram Protocol (UDP) is called a connectionless, unreliable transport protocol. It does not add anything to the services of IP except to provide process-to-process communication instead of host-to-host communication.

Topics discussed in this section:

Well-Known Ports for UDP

User Datagram

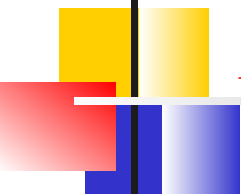
Checksum

UDP Operation

Use of UDP

Table 23.1 *Well-known ports used with UDP*

<i>Port</i>	<i>Protocol</i>	<i>Description</i>
7	Echo	Echoes a received datagram back to the sender
9	Discard	Discards any datagram that is received
11	Users	Active users
13	Daytime	Returns the date and the time
17	Quote	Returns a quote of the day
19	Chargen	Returns a string of characters
53	Nameserver	Domain Name Service
67	BOOTPs	Server port to download bootstrap information
68	BOOTPc	Client port to download bootstrap information
69	TFTP	Trivial File Transfer Protocol
111	RPC	Remote Procedure Call
123	NTP	Network Time Protocol
161	SNMP	Simple Network Management Protocol
162	SNMP	Simple Network Management Protocol (trap)



Example 23.1

In UNIX, the well-known ports are stored in a file called `/etc/services`. Each line in this file gives the name of the server and the well-known port number. We can use the `grep` utility to extract the line corresponding to the desired application. The following shows the port for FTP. Note that FTP can use port 21 with either UDP or TCP.

```
$ grep ftp /etc/services
ftp      21/tcp
ftp      21/udp
```

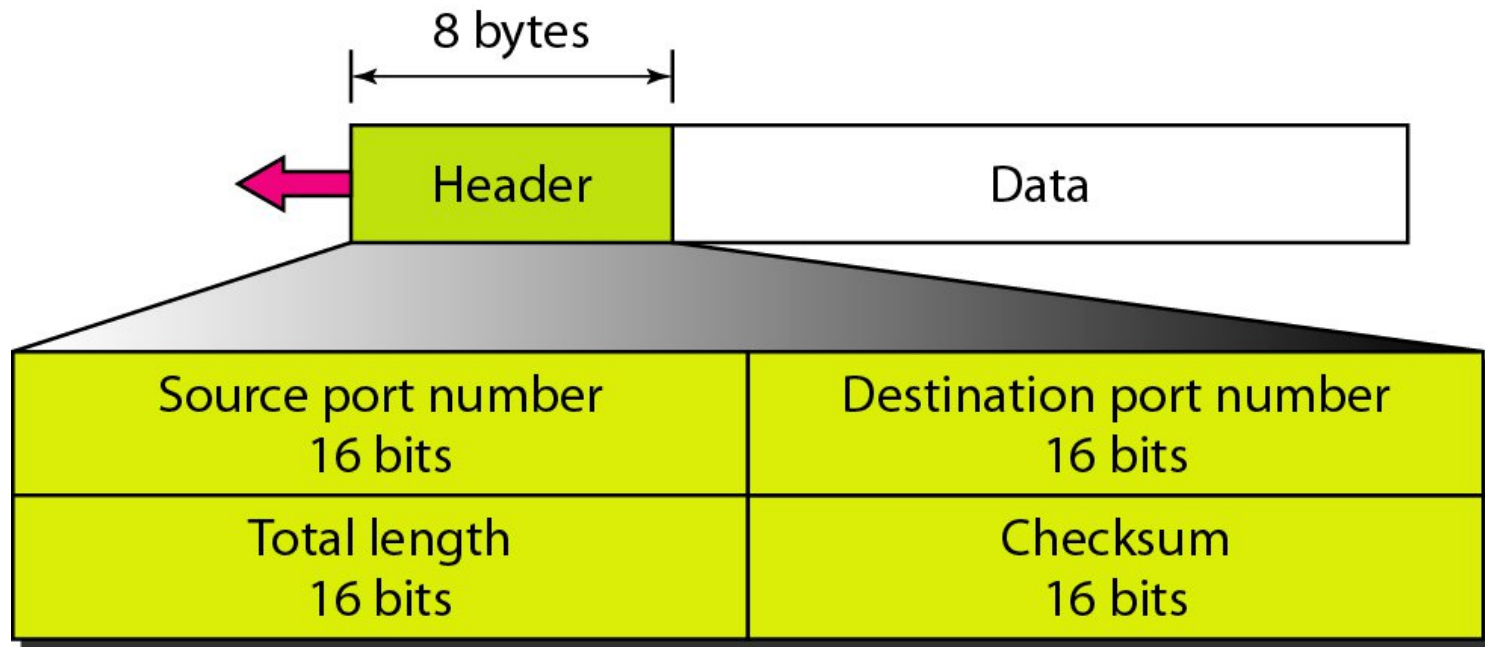


Example 23.1 (continued)

SNMP uses two port numbers (161 and 162), each for a different purpose, as we will see in Chapter 28.

```
$ grep      snmp /etc/services
snmp        161/tcp      #Simple Net  Mgmt Proto
snmp        161/udp      #Simple Net  Mgmt Proto
snmptrap    162/udp      #Traps for SNMP
```

Figure 23.9 *User datagram format*

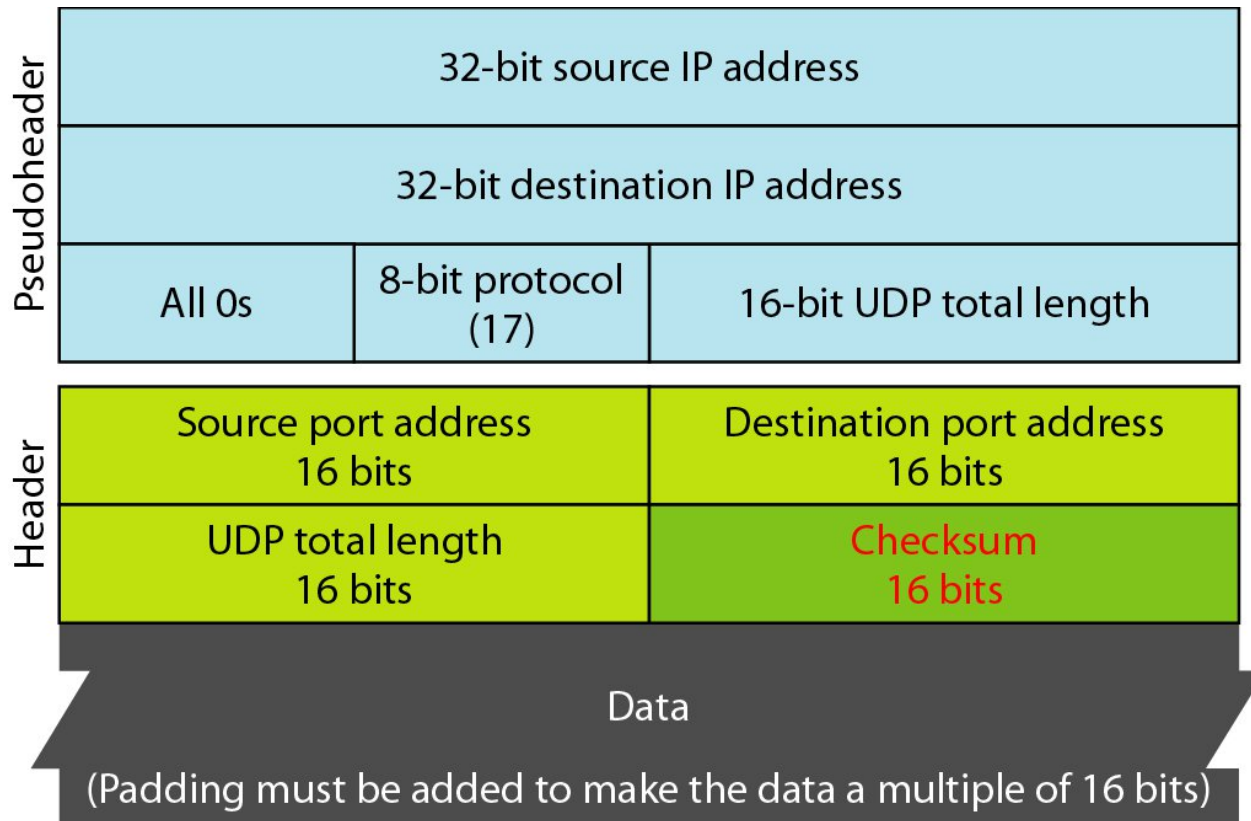




Note

**UDP length
= IP length – IP header's length**

Figure 23.10 *Pseudoheader for checksum calculation*





Example 23.2

Figure 23.11 shows the checksum calculation for a very small user datagram with only 7 bytes of data. Because the number of bytes of data is odd, padding is added for checksum calculation. The pseudoheader as well as the padding will be dropped when the user datagram is delivered to IP.

Figure 23.11 *Checksum calculation of a simple UDP user datagram*

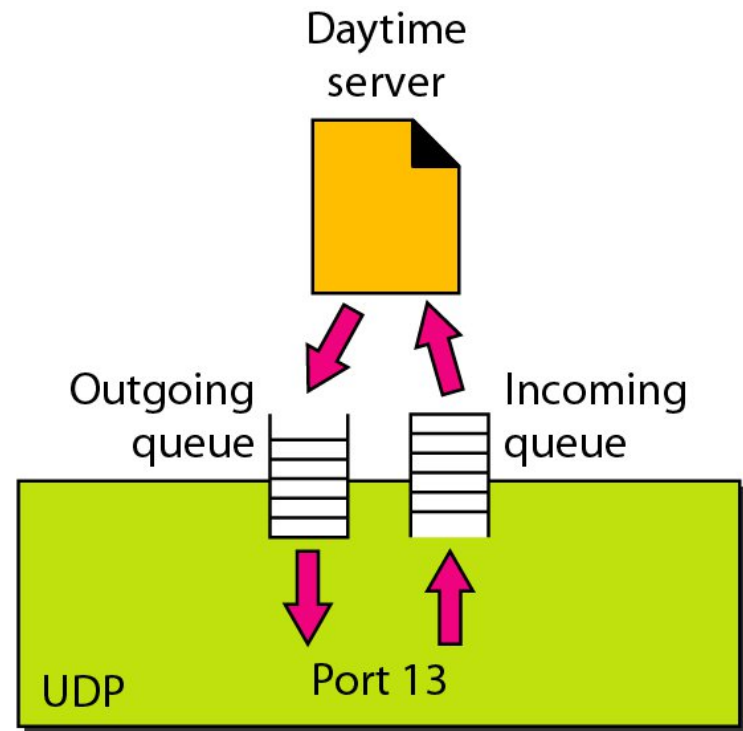
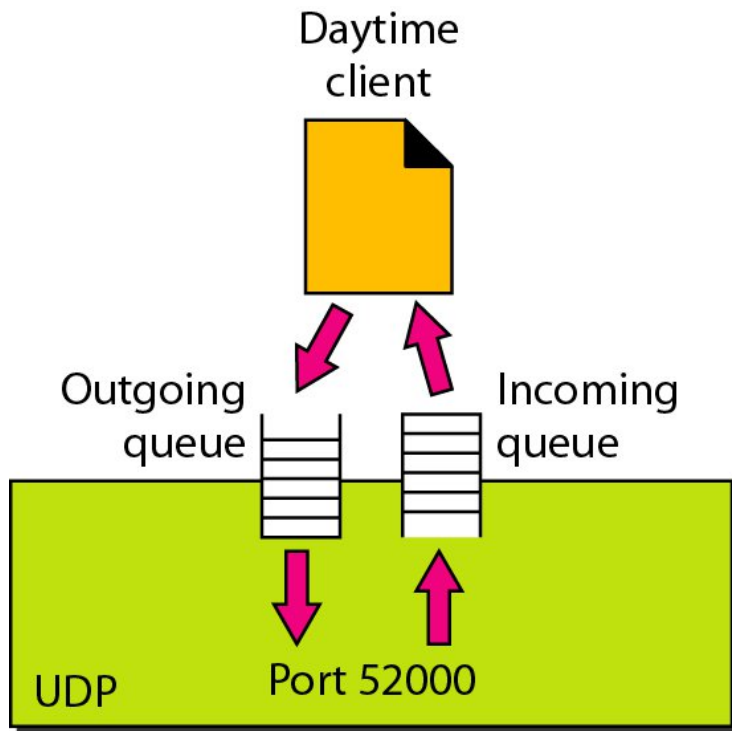
153.18.8.105		
171.2.14.10		
All 0s	17	15

1087	13
15	All 0s

T	E	S	T
I	N	G	All 0s

10011001	00010010	→	153.18
00001000	01101001	→	8.105
10101011	00000010	→	171.2
00001110	00001010	→	14.10
00000000	00010001	→	0 and 17
00000000	00001111	→	15
00000100	00111111	→	1087
00000000	00001101	→	13
00000000	00001111	→	15
00000000	00000000	→	0 (checksum)
01010100	01000101	→	T and E
01010011	01010100	→	S and T
01001001	01001110	→	I and N
01000111	00000000	→	G and 0 (padding)
10010110	11101011	→	Sum
01101001	00010100	→	Checksum

Figure 23.12 *Queues in UDP*



23-3 TCP

TCP is a connection-oriented protocol; it creates a virtual connection between two TCPs to send data. In addition, TCP uses flow and error control mechanisms at the transport level.

Topics discussed in this section:

TCP Services

TCP Features

Segment

A TCP Connection

Flow Control

Error Control

Table 23.2 *Well-known ports used by TCP*

<i>Port</i>	<i>Protocol</i>	<i>Description</i>
7	Echo	Echoes a received datagram back to the sender
9	Discard	Discards any datagram that is received
11	Users	Active users
13	Daytime	Returns the date and the time
17	Quote	Returns a quote of the day
19	Chargen	Returns a string of characters
20	FTP, Data	File Transfer Protocol (data connection)
21	FTP, Control	File Transfer Protocol (control connection)
23	TELNET	Terminal Network
25	SMTP	Simple Mail Transfer Protocol
53	DNS	Domain Name Server
67	BOOTP	Bootstrap Protocol
79	Finger	Finger
80	HTTP	Hypertext Transfer Protocol
111	RPC	Remote Procedure Call

Figure 23.13 *Stream delivery*

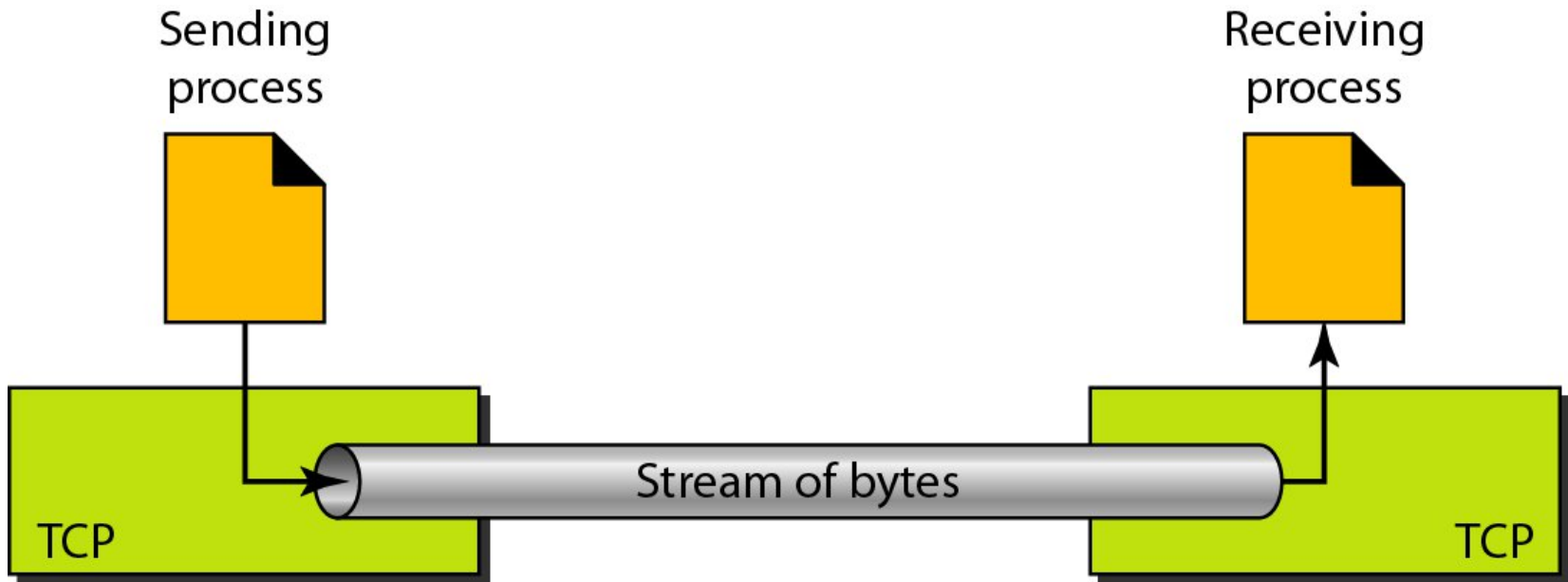


Figure 23.14 *Sending and receiving buffers*

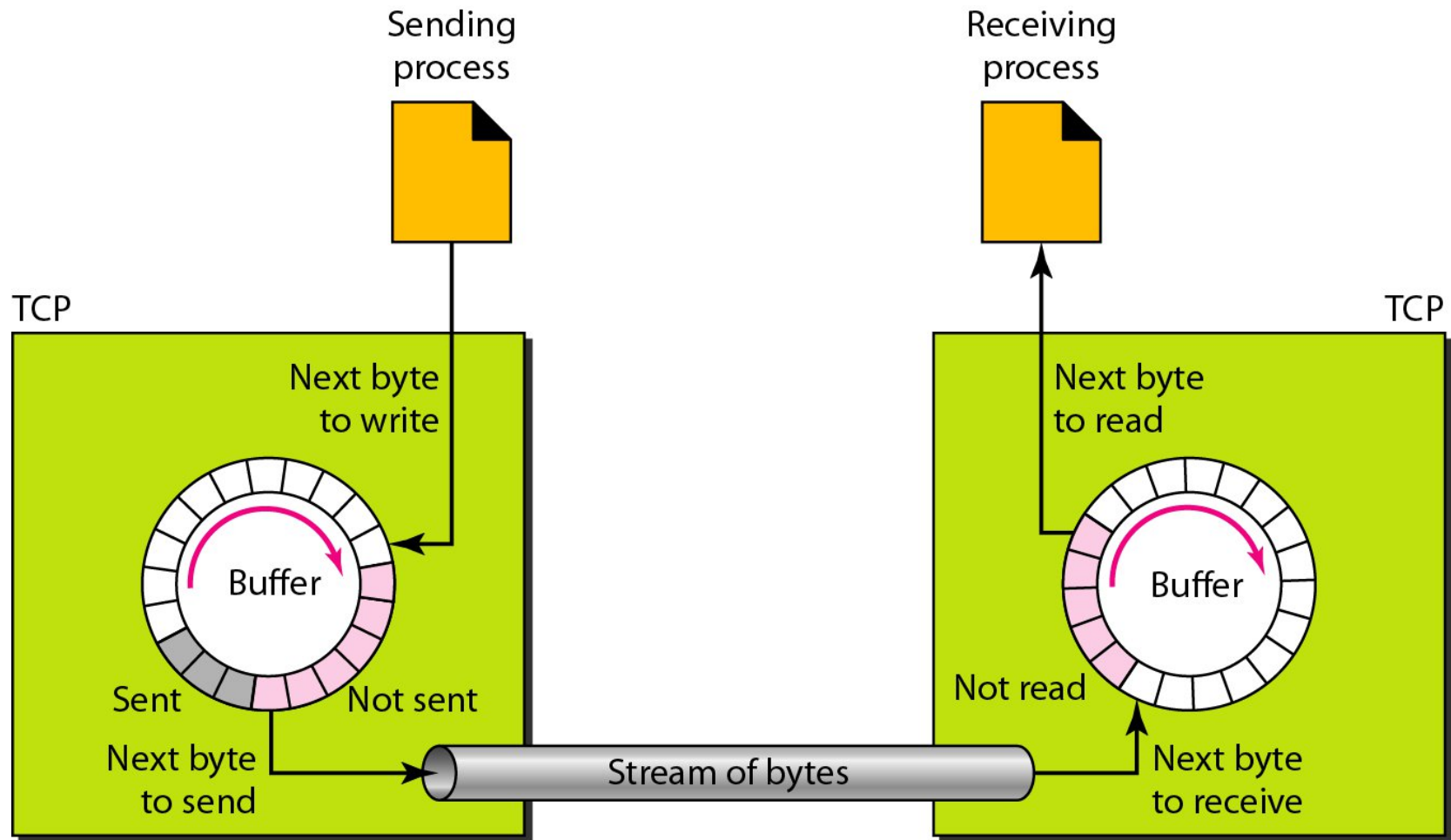
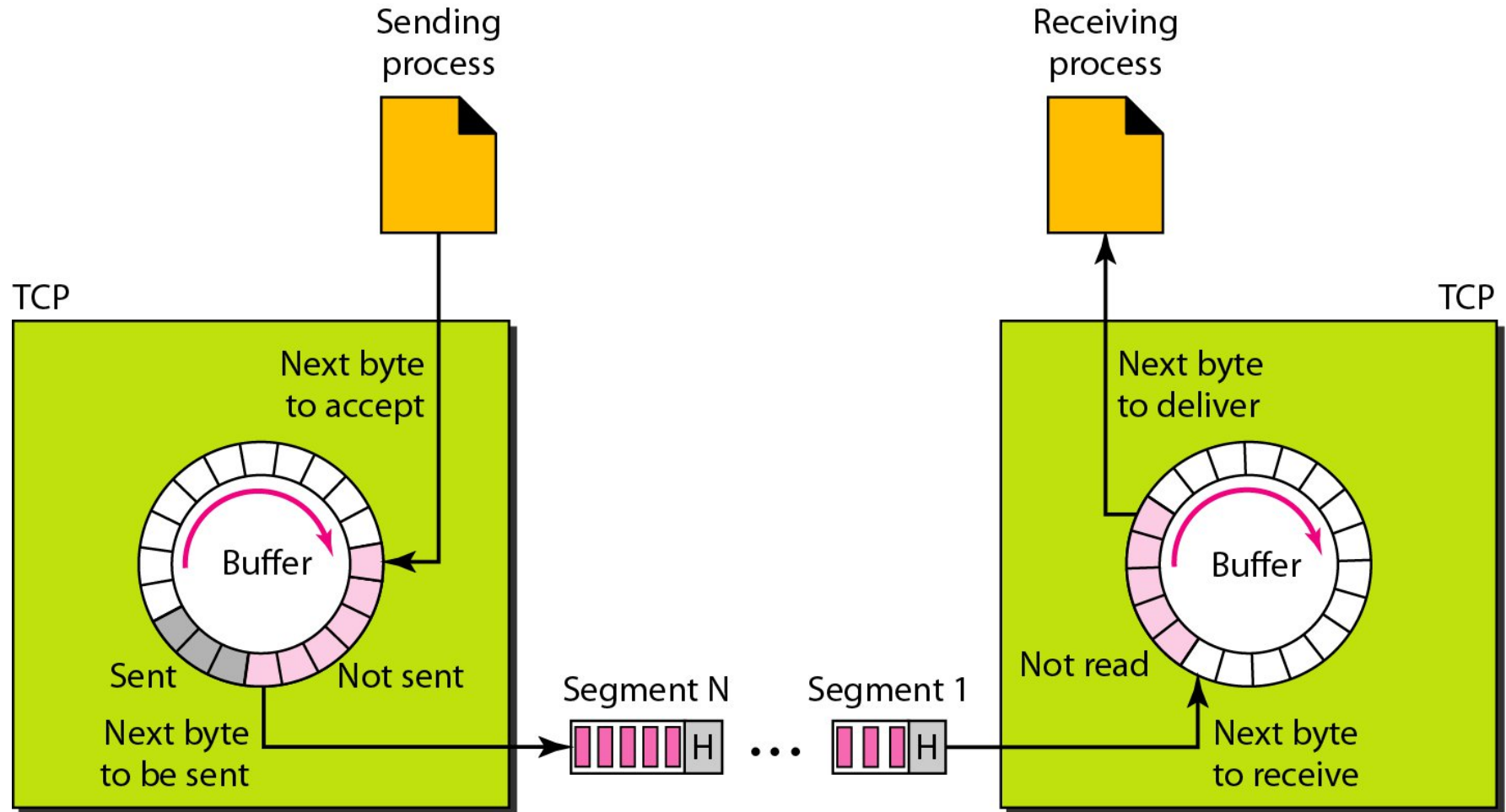


Figure 23.15 *TCP segments*



Note

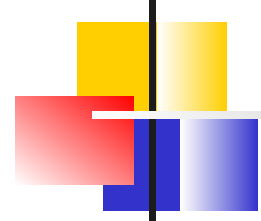
The bytes of data being transferred in each connection are numbered by TCP. The numbering starts with a randomly generated number.



Example 23.3

The following shows the sequence number for each segment:

Segment 1	➡	Sequence Number: 10,001 (range: 10,001 to 11,000)
Segment 2	➡	Sequence Number: 11,001 (range: 11,001 to 12,000)
Segment 3	➡	Sequence Number: 12,001 (range: 12,001 to 13,000)
Segment 4	➡	Sequence Number: 13,001 (range: 13,001 to 14,000)
Segment 5	➡	Sequence Number: 14,001 (range: 14,001 to 15,000)



Note

The value in the sequence number field of a segment defines the number of the first data byte contained in that segment.

Note

**The value of the acknowledgment field
in a segment defines
the number of the next byte a party
expects to receive.
The acknowledgment number is
cumulative.**

Figure 23.16 *TCP segment format*

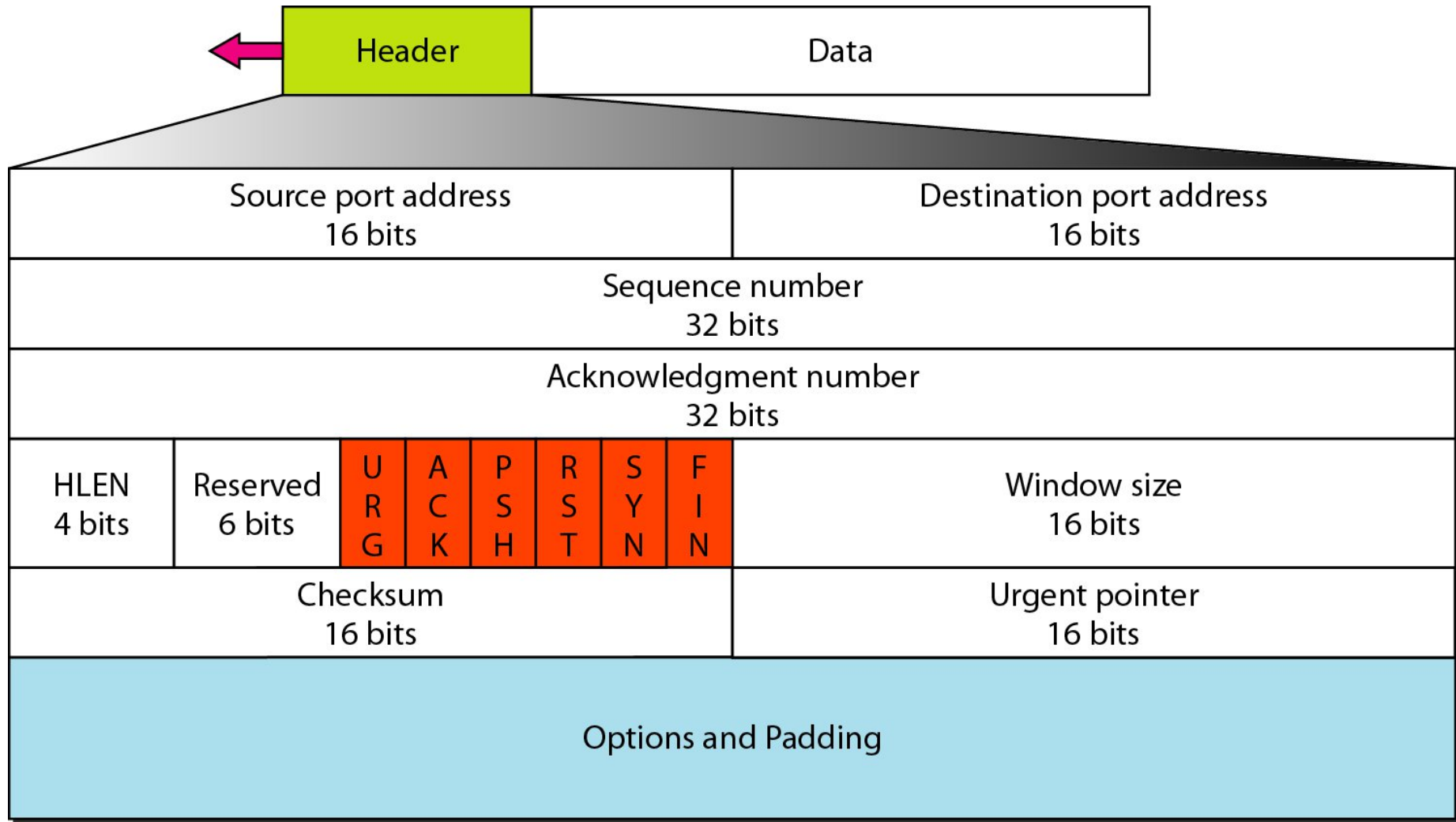


Figure 23.17 *Control field*

URG: Urgent pointer is valid
ACK: Acknowledgment is valid
PSH: Request for push

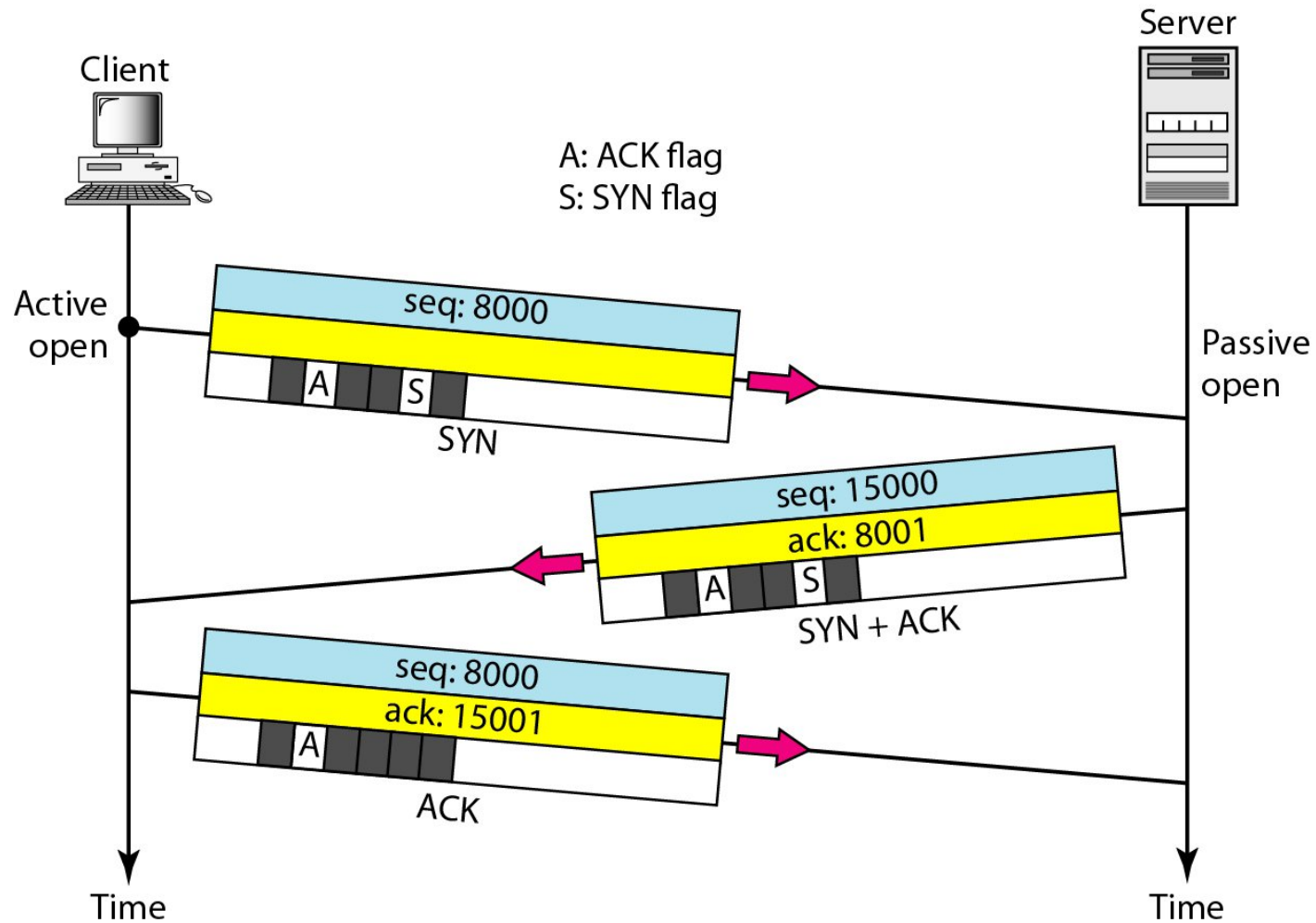
RST: Reset the connection
SYN: Synchronize sequence numbers
FIN: Terminate the connection



Table 23.3 *Description of flags in the control field*

<i>Flag</i>	<i>Description</i>
URG	The value of the urgent pointer field is valid.
ACK	The value of the acknowledgment field is valid.
PSH	Push the data.
RST	Reset the connection.
SYN	Synchronize sequence numbers during connection.
FIN	Terminate the connection.

Figure 23.18 *Connection establishment using three-way handshaking*





A SYN segment cannot carry data, but it consumes one sequence number.

Note

A SYN + ACK segment cannot carry data, but does consume one sequence number.



**An ACK segment, if carrying no data,
consumes no sequence number.**

Figure 23.19 *Data transfer*

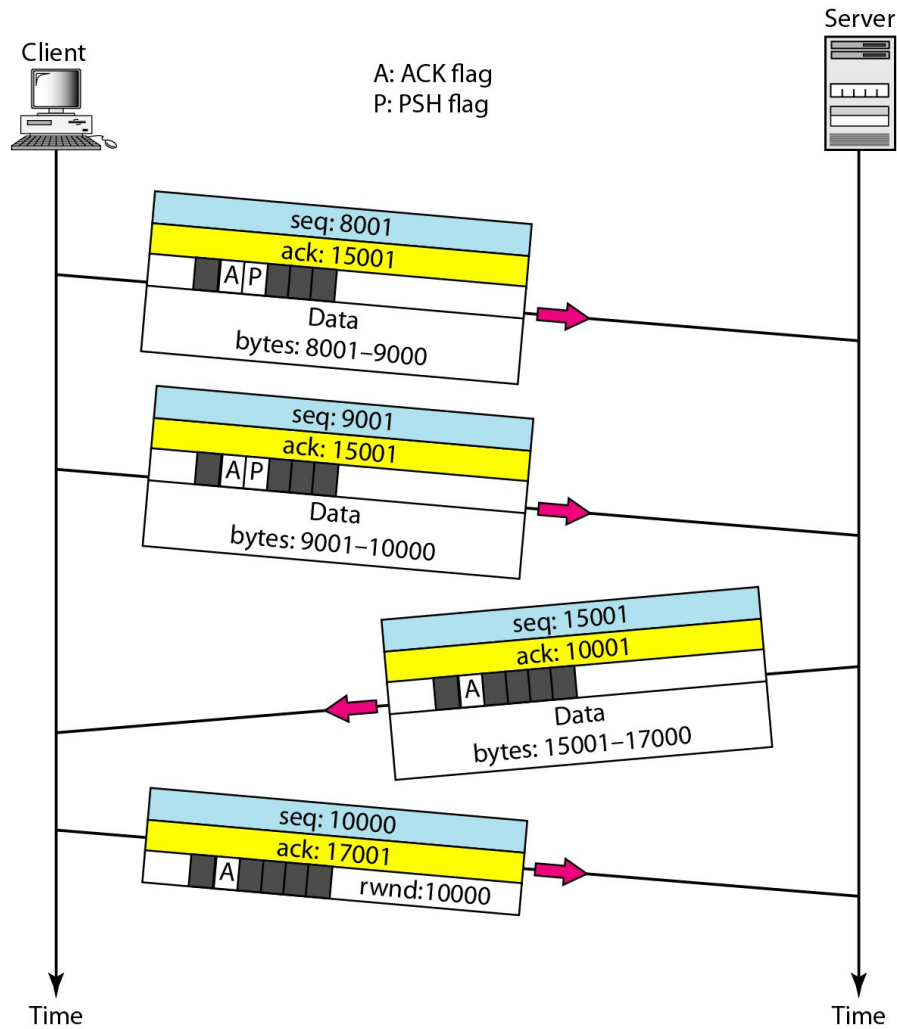
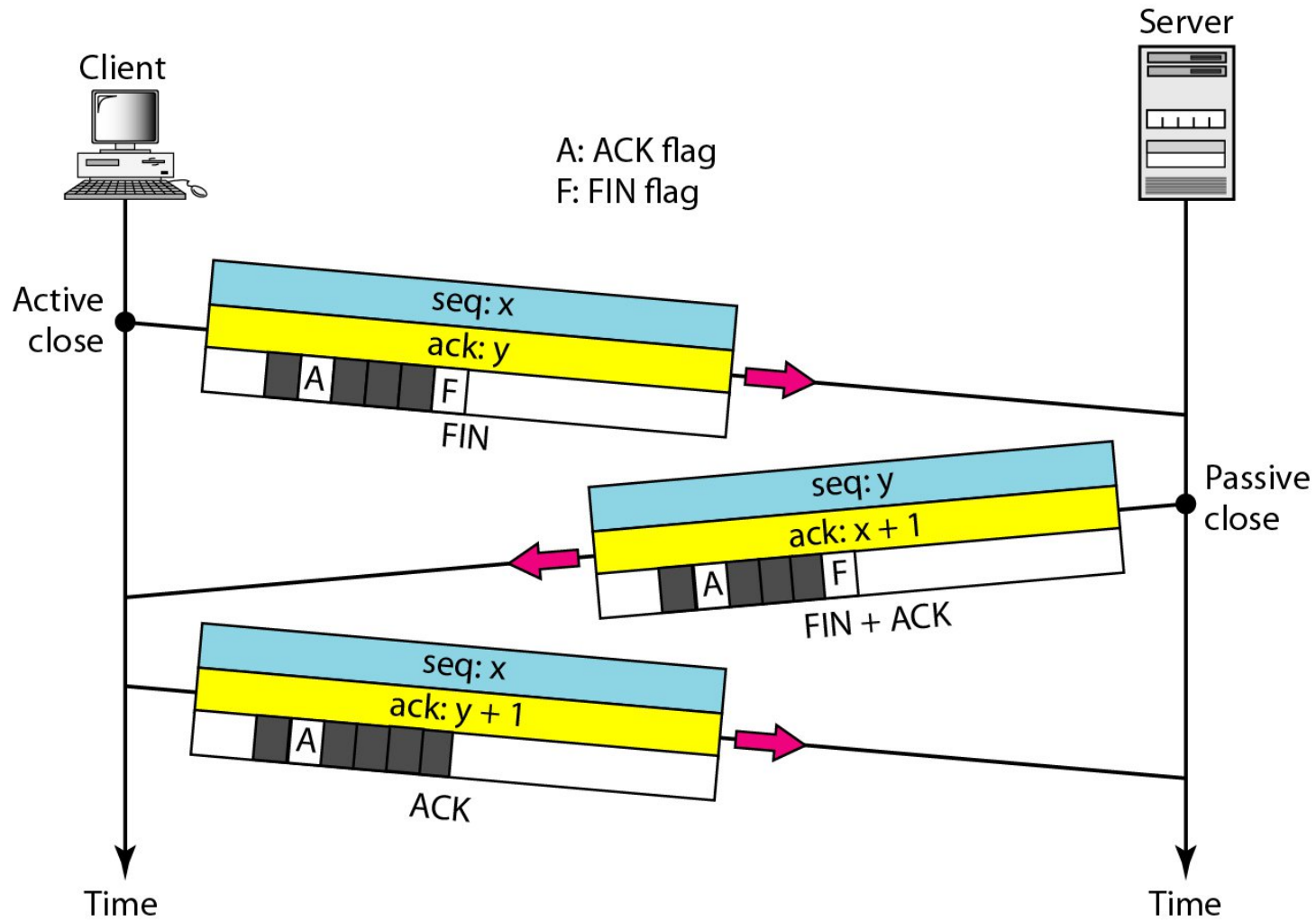


Figure 23.20 *Connection termination using three-way handshaking*



Note

The FIN segment consumes one sequence number if it does not carry data.

Note

**The FIN + ACK segment consumes
one sequence number if it
does not carry data.**

Figure 23.21 *Half-close*

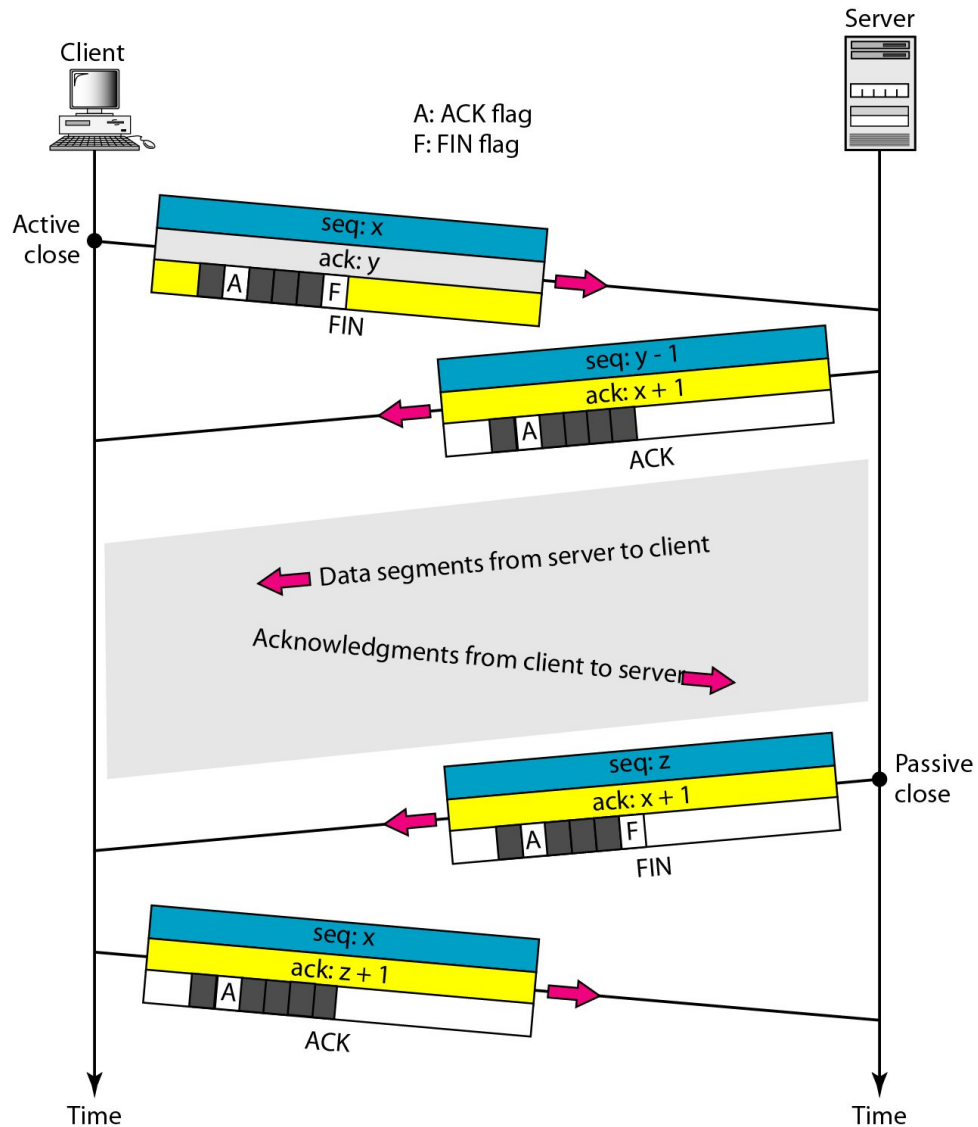
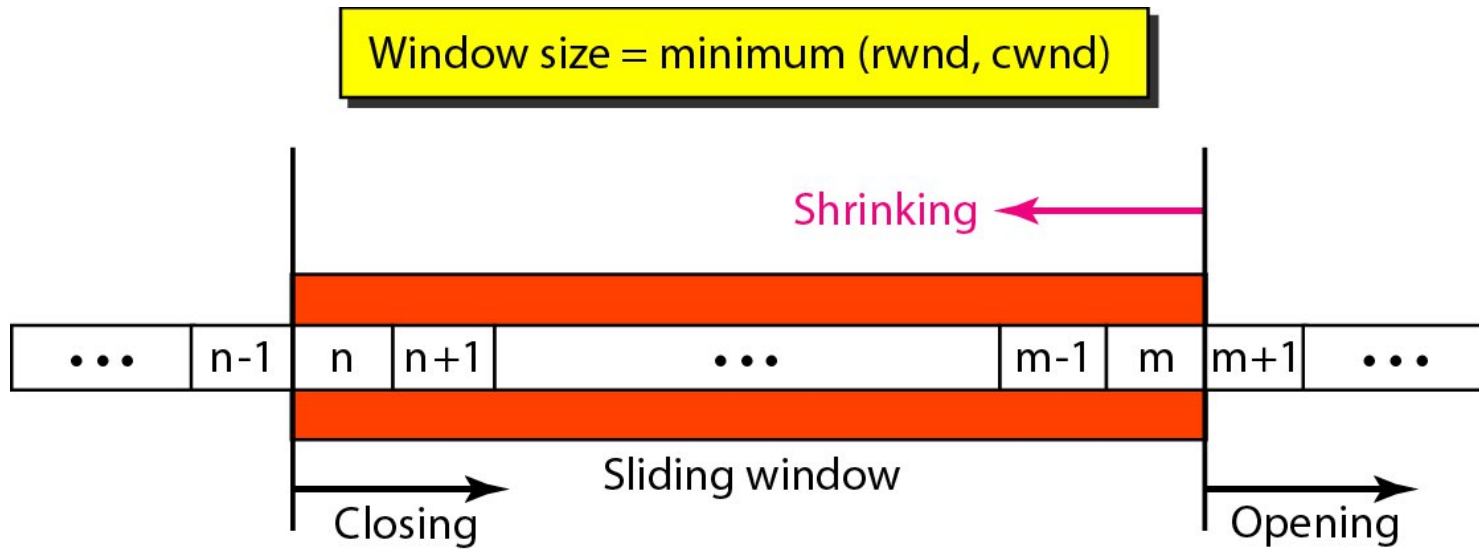


Figure 23.22 *Sliding window*



Note

**A sliding window is used to make transmission more efficient as well as to control the flow of data so that the destination does not become overwhelmed with data.
TCP sliding windows are byte-oriented.**



Example 23.4

What is the value of the receiver window (rwnd) for host A if the receiver, host B, has a buffer size of 5000 bytes and 1000 bytes of received and unprocessed data?

Solution

The value of $rwnd = 5000 - 1000 = 4000$. Host B can receive only 4000 bytes of data before overflowing its buffer. Host B advertises this value in its next segment to A.



Example 23.5

What is the size of the window for host A if the value of $rwnd$ is 3000 bytes and the value of $cwnd$ is 3500 bytes?

Solution

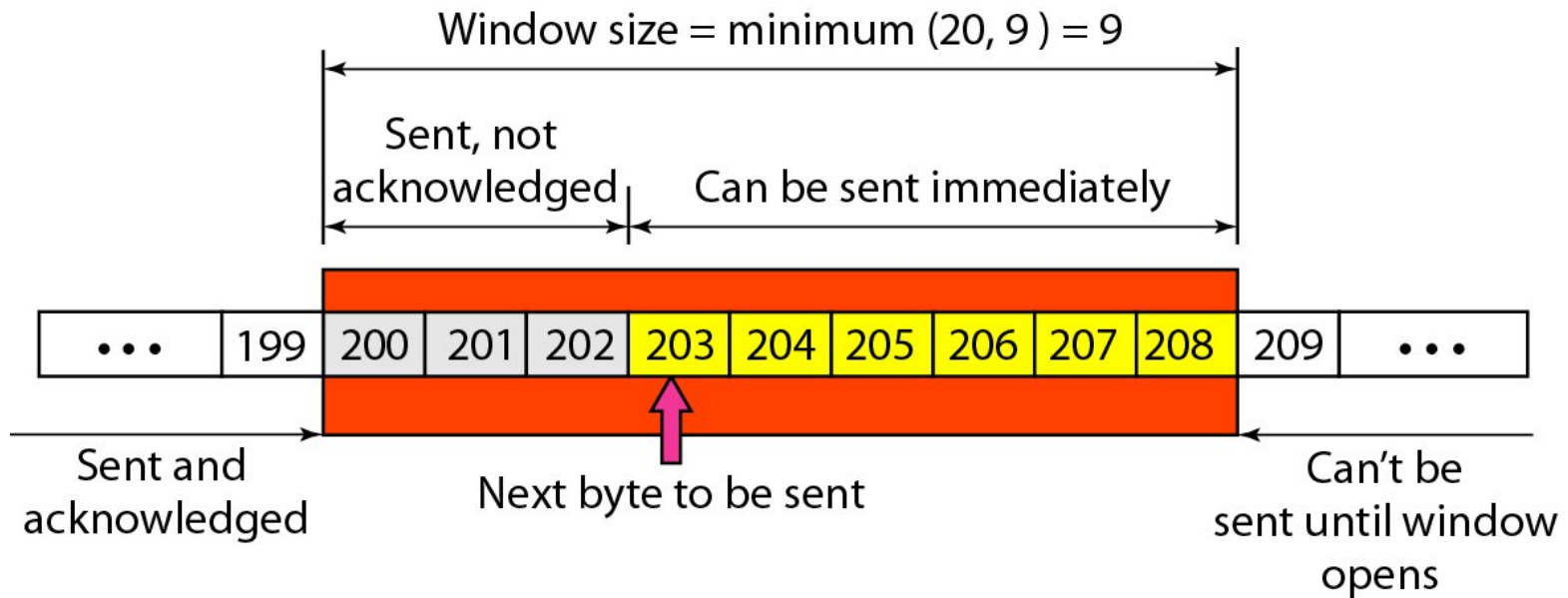
The size of the window is the smaller of $rwnd$ and $cwnd$, which is 3000 bytes.



Example 23.6

Figure 23.23 shows an unrealistic example of a sliding window. The sender has sent bytes up to 202. We assume that $cwnd$ is 20 (in reality this value is thousands of bytes). The receiver has sent an acknowledgment number of 200 with an $rwnd$ of 9 bytes (in reality this value is thousands of bytes). The size of the sender window is the minimum of $rwnd$ and $cwnd$, or 9 bytes. Bytes 200 to 202 are sent, but not acknowledged. Bytes 203 to 208 can be sent without worrying about acknowledgment. Bytes 209 and above cannot be sent.

Figure 23.23 *Example 23.6*



Some points about TCP sliding windows:

- ❑ The size of the window is the lesser of $rwnd$ and $cwnd$.
- ❑ The source does not have to send a full window's worth of data.
- ❑ The window can be opened or closed by the receiver, but should not be shrunk.
- ❑ The destination can send an acknowledgment at any time as long as it does not result in a shrinking window.
- ❑ The receiver can temporarily shut down the window; the sender, however, can always send a segment of 1 byte after the window is shut down.



Note

**ACK segments do not consume
sequence numbers and are not
acknowledged.**

Note

In modern implementations, a retransmission occurs if the retransmission timer expires or three duplicate ACK segments have arrived.



No retransmission timer is set for an ACK segment.

Note

Data may arrive out of order and be temporarily stored by the receiving TCP, but TCP guarantees that no out-of-order segment is delivered to the process.

Figure 23.24 *Normal operation*

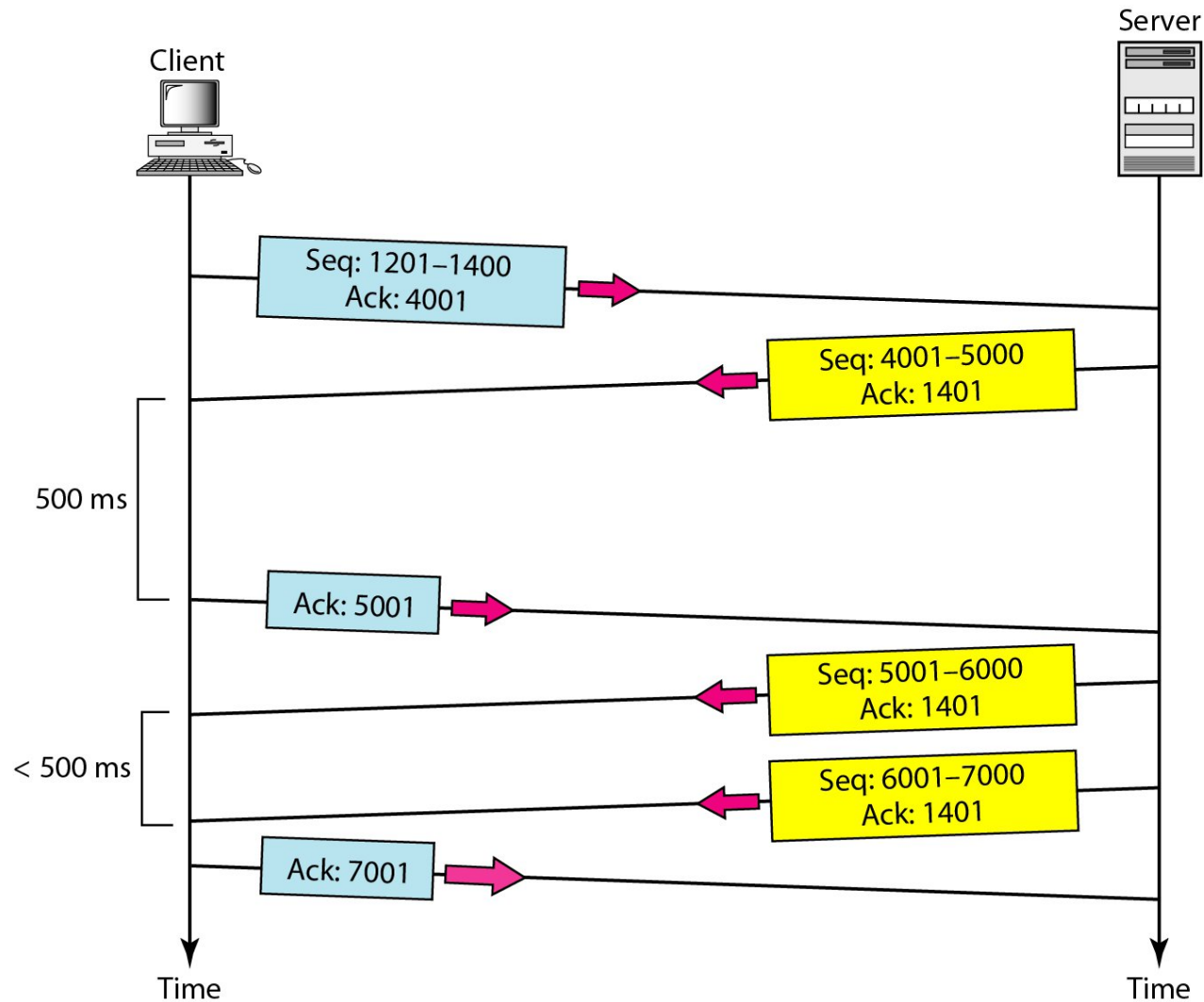
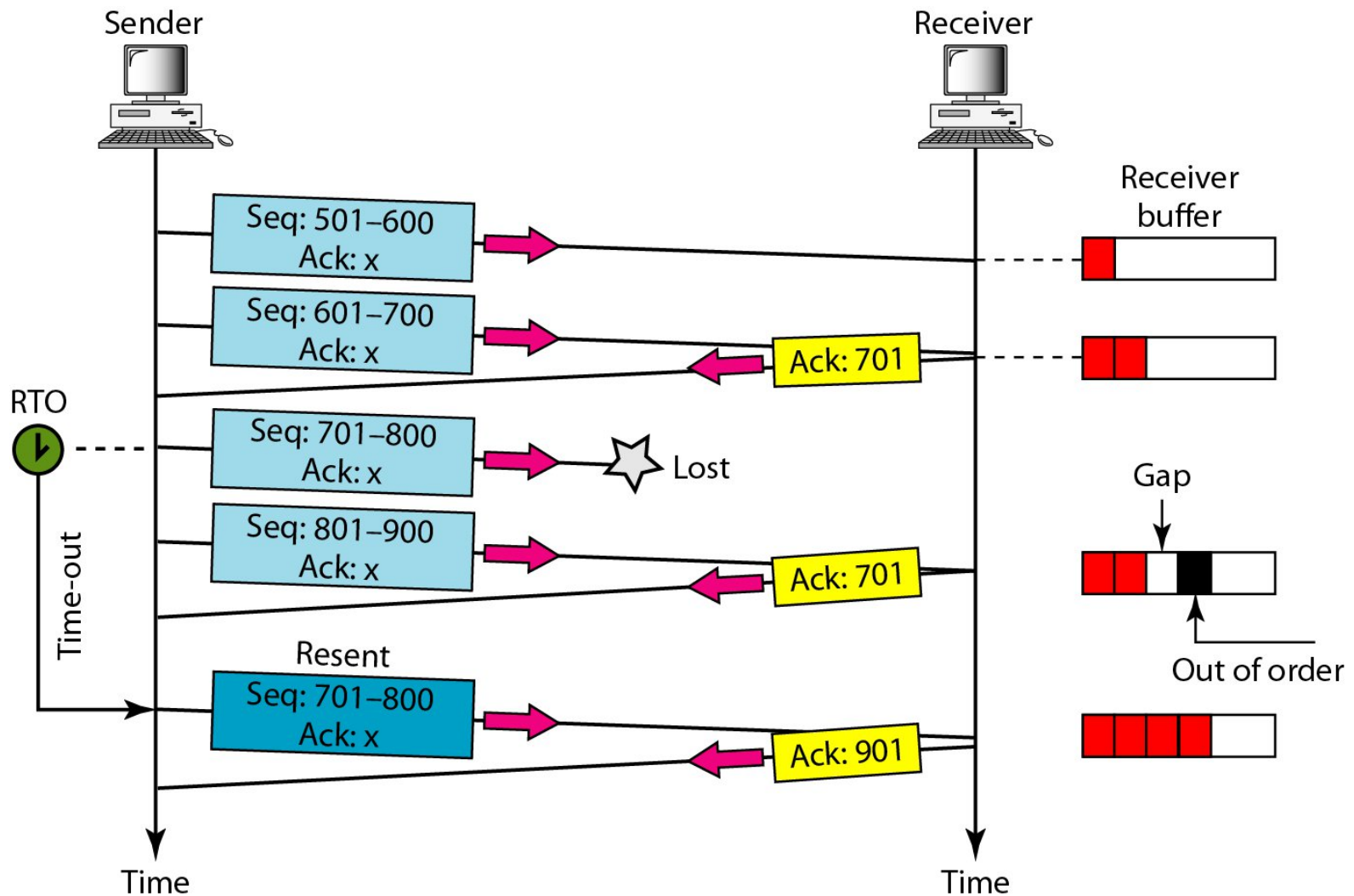


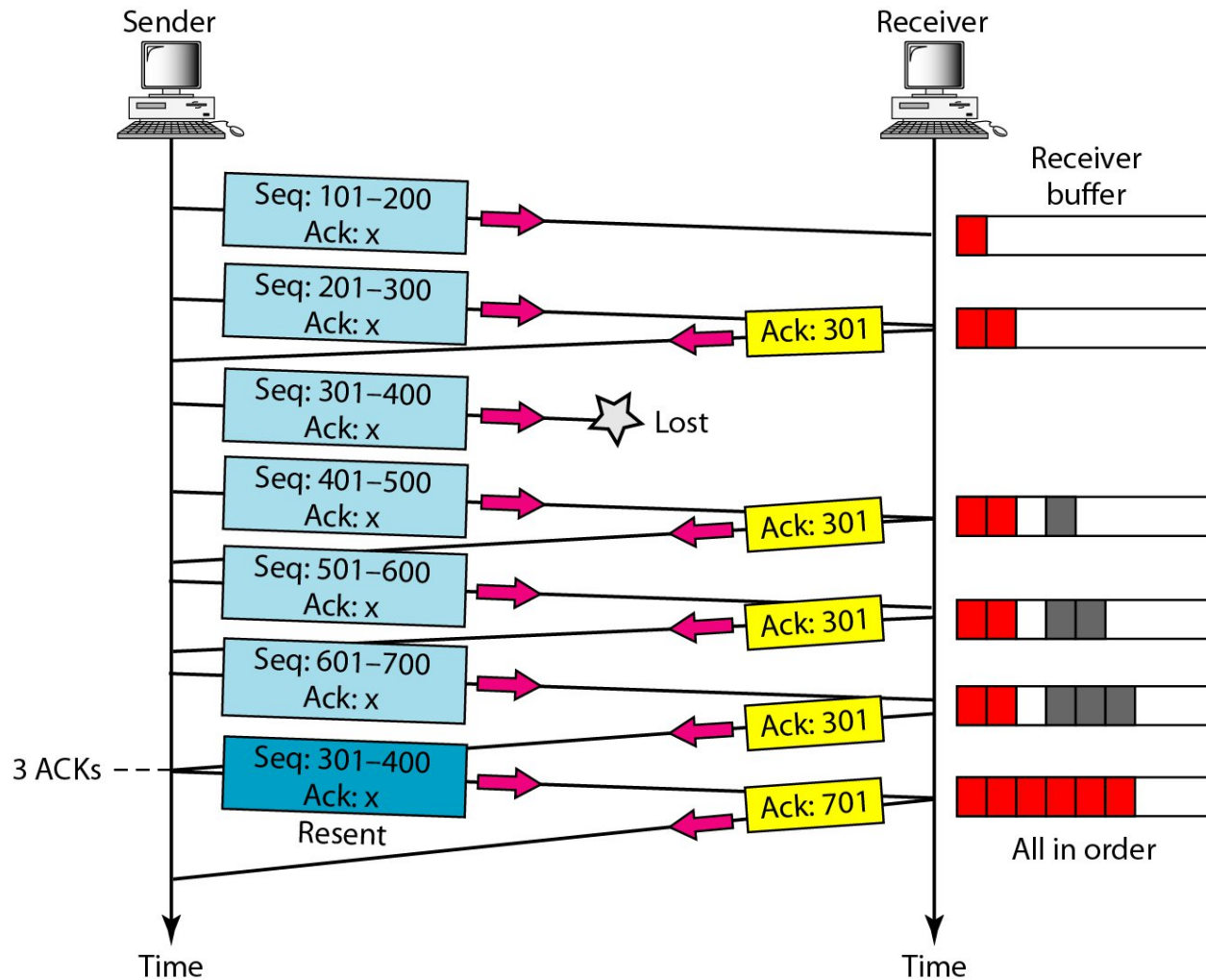
Figure 23.25 *Lost segment*





The receiver TCP delivers only ordered data to the process.

Figure 23.26 *Fast retransmission*



23-4 SCTP

Stream Control Transmission Protocol (SCTP) is a new reliable, message-oriented transport layer protocol. SCTP, however, is mostly designed for Internet applications that have recently been introduced. These new applications need a more sophisticated service than TCP can provide.

Topics discussed in this section:

SCTP Services and Features

Packet Format

An SCTP Association

Flow Control and Error Control

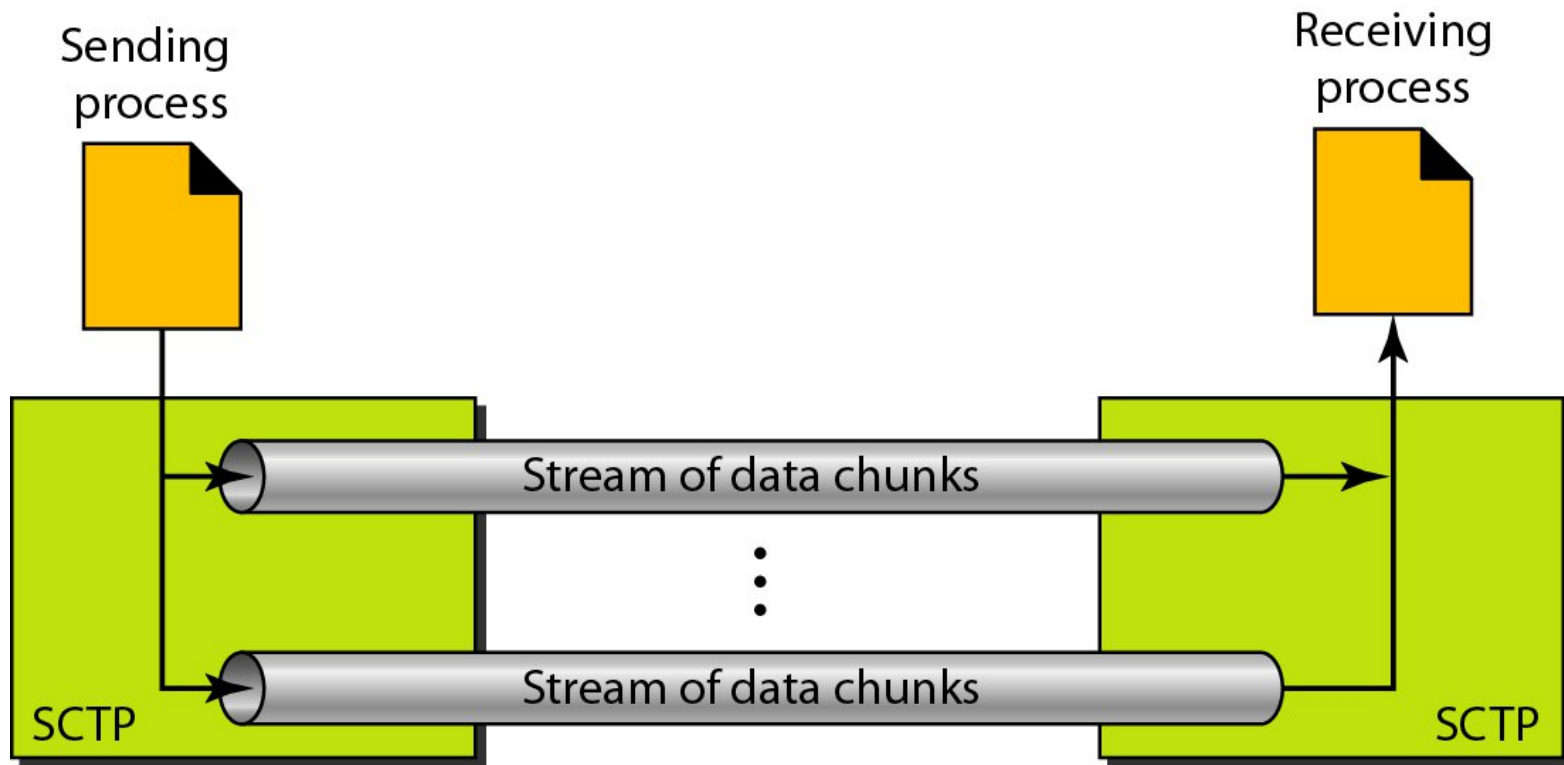


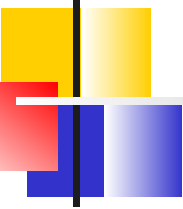
SCTP is a message-oriented, reliable protocol that combines the best features of UDP and TCP.

Table 23.4 *Some SCTP applications*

<i>Protocol</i>	<i>Port Number</i>	<i>Description</i>
IUA	9990	ISDN over IP
M2UA	2904	SS7 telephony signaling
M3UA	2905	SS7 telephony signaling
H.248	2945	Media gateway control
H.323	1718, 1719, 1720, 11720	IP telephony
SIP	5060	IP telephony

Figure 23.27 *Multiple-stream concept*

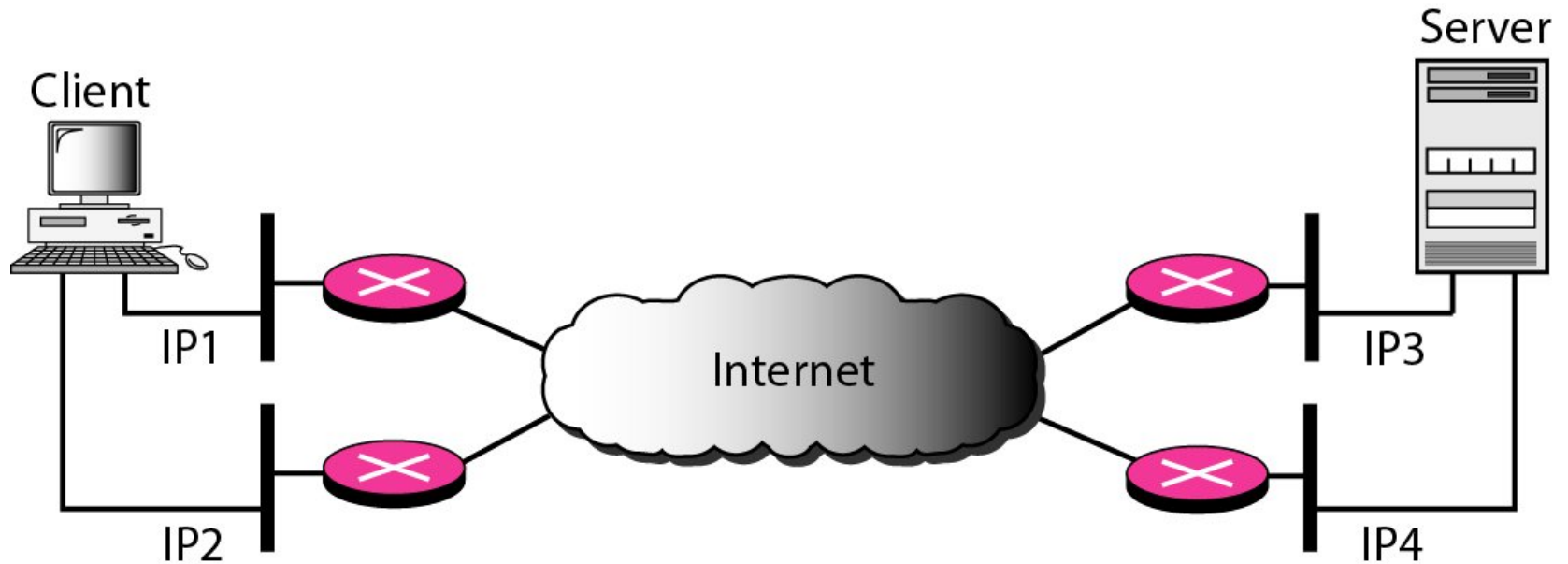




Note

An association in SCTP can involve multiple streams.

Figure 23.28 *Multihoming concept*





SCTP association allows multiple IP addresses for each end.



In SCTP, a data chunk is numbered using a TSN.



To distinguish between different streams, SCTP uses an SI.



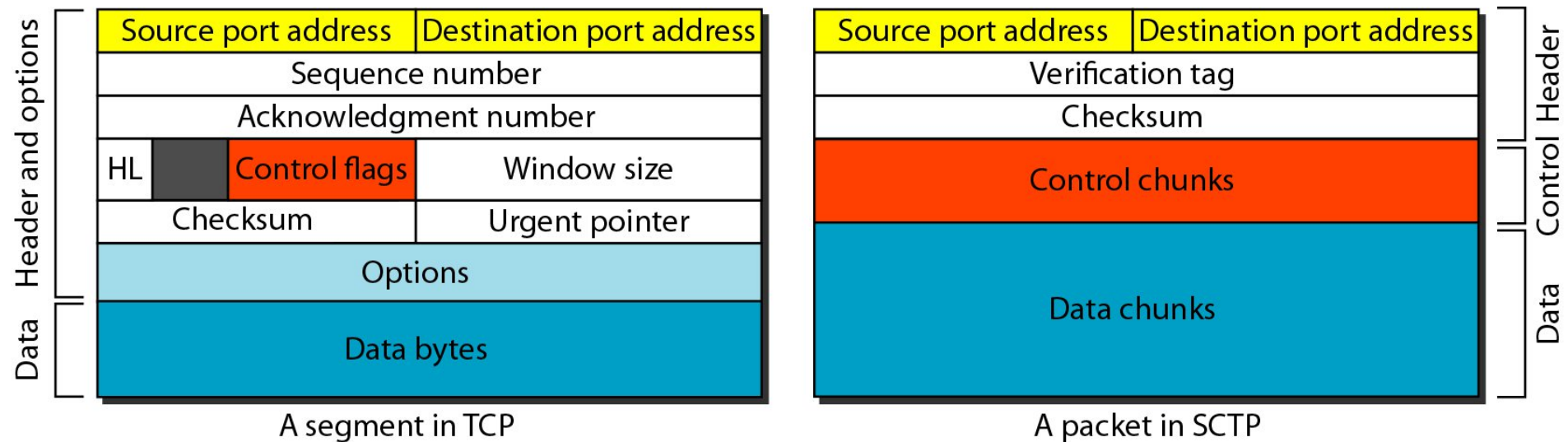
Note

To distinguish between different data chunks belonging to the same stream, SCTP uses SSNs.



TCP has segments; SCTP has packets.

Figure 23.29 *Comparison between a TCP segment and an SCTP packet*

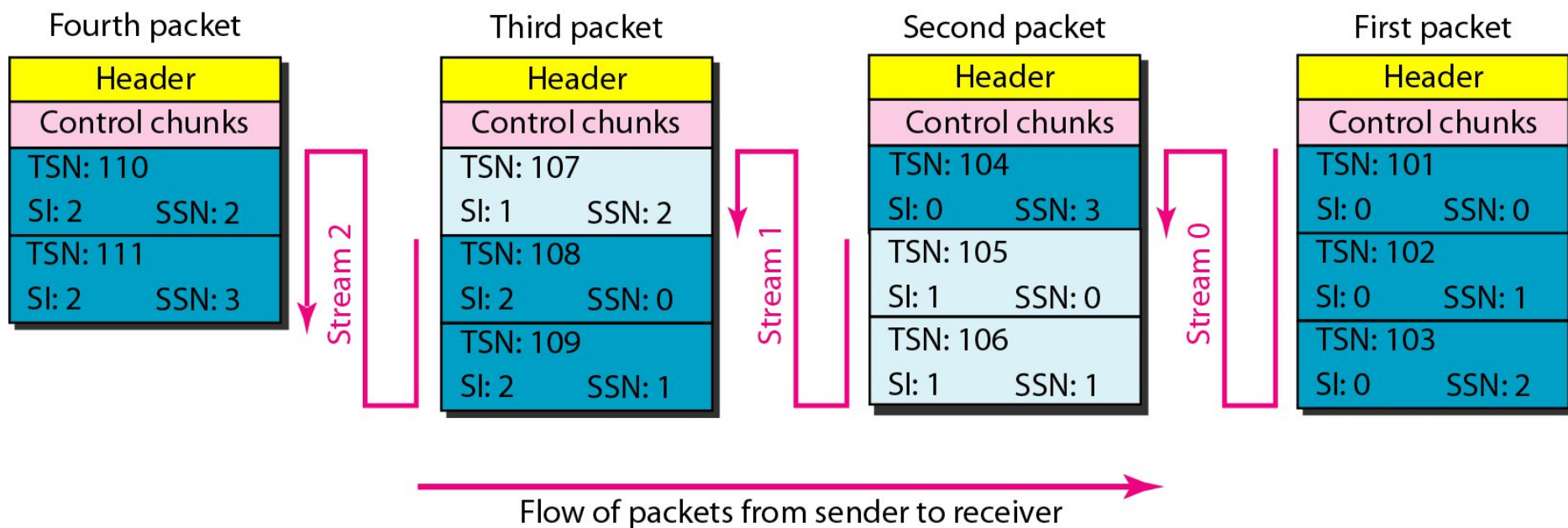




Note

In SCTP, control information and data information are carried in separate chunks.

Figure 23.30 *Packet, data chunks, and streams*





Note

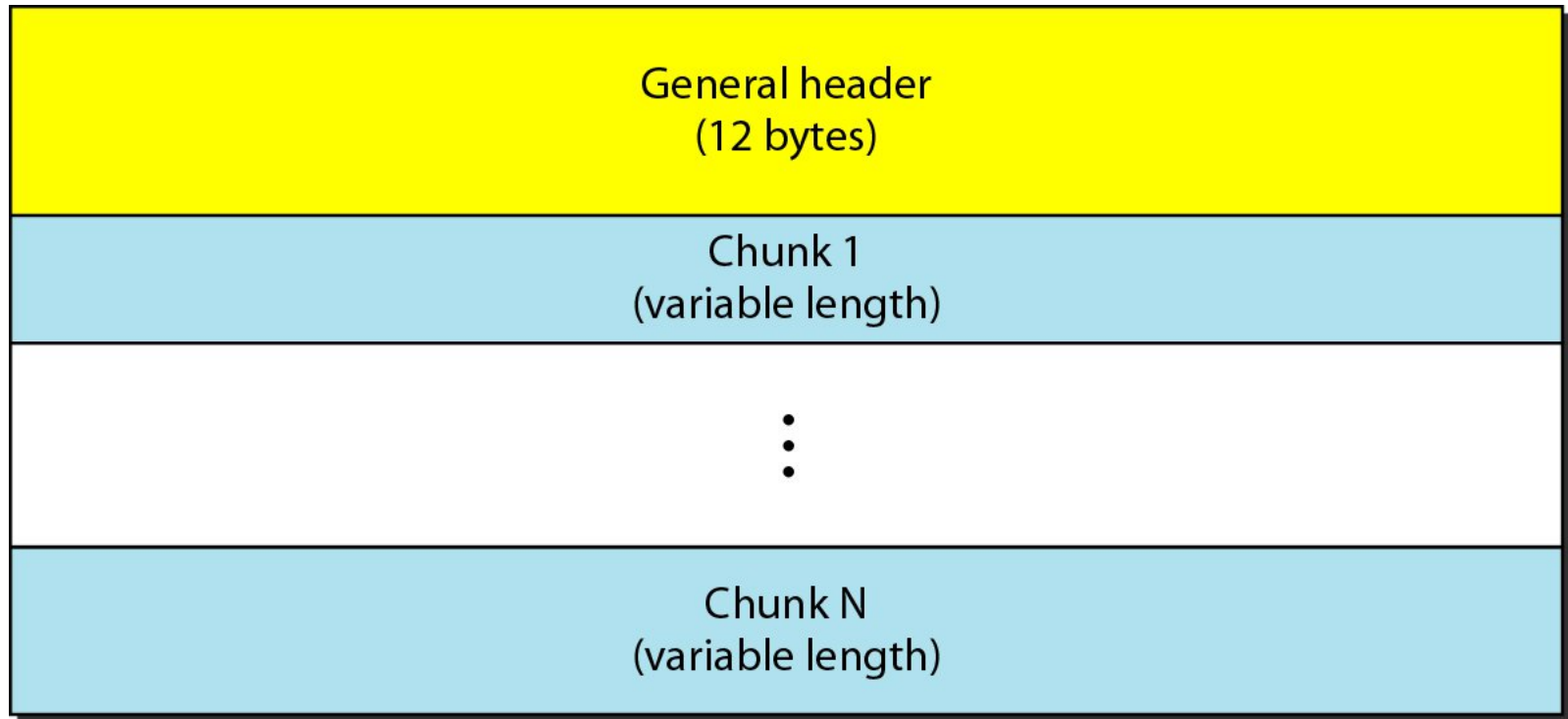
Data chunks are identified by three items: TSN, SI, and SSN.

TSN is a cumulative number identifying the association; SI defines the stream; SSN defines the chunk in a stream.

Note

In SCTP, acknowledgment numbers are used to acknowledge only data chunks; control chunks are acknowledged by other control chunks if necessary.

Figure 23.31 *SCTP packet format*





In an SCTP packet, control chunks come before data chunks.

Figure 23.32 *General header*

Source port address 16 bits	Destination port address 16 bits
Verification tag 32 bits	
Checksum 32 bits	

Table 23.5 *Chunks*

<i>Type</i>	<i>Chunk</i>	<i>Description</i>
0	DATA	User data
1	INIT	Sets up an association
2	INIT ACK	Acknowledges INIT chunk
3	SACK	Selective acknowledgment
4	HEARTBEAT	Probes the peer for liveliness
5	HEARTBEAT ACK	Acknowledges HEARTBEAT chunk
6	ABORT	Aborts an association
7	SHUTDOWN	Terminates an association
8	SHUTDOWN ACK	Acknowledges SHUTDOWN chunk
9	ERROR	Reports errors without shutting down
10	COOKIE ECHO	Third packet in association establishment
11	COOKIE ACK	Acknowledges COOKIE ECHO chunk
14	SHUTDOWN COMPLETE	Third packet in association termination
192	FORWARD TSN	For adjusting cumulative TSN

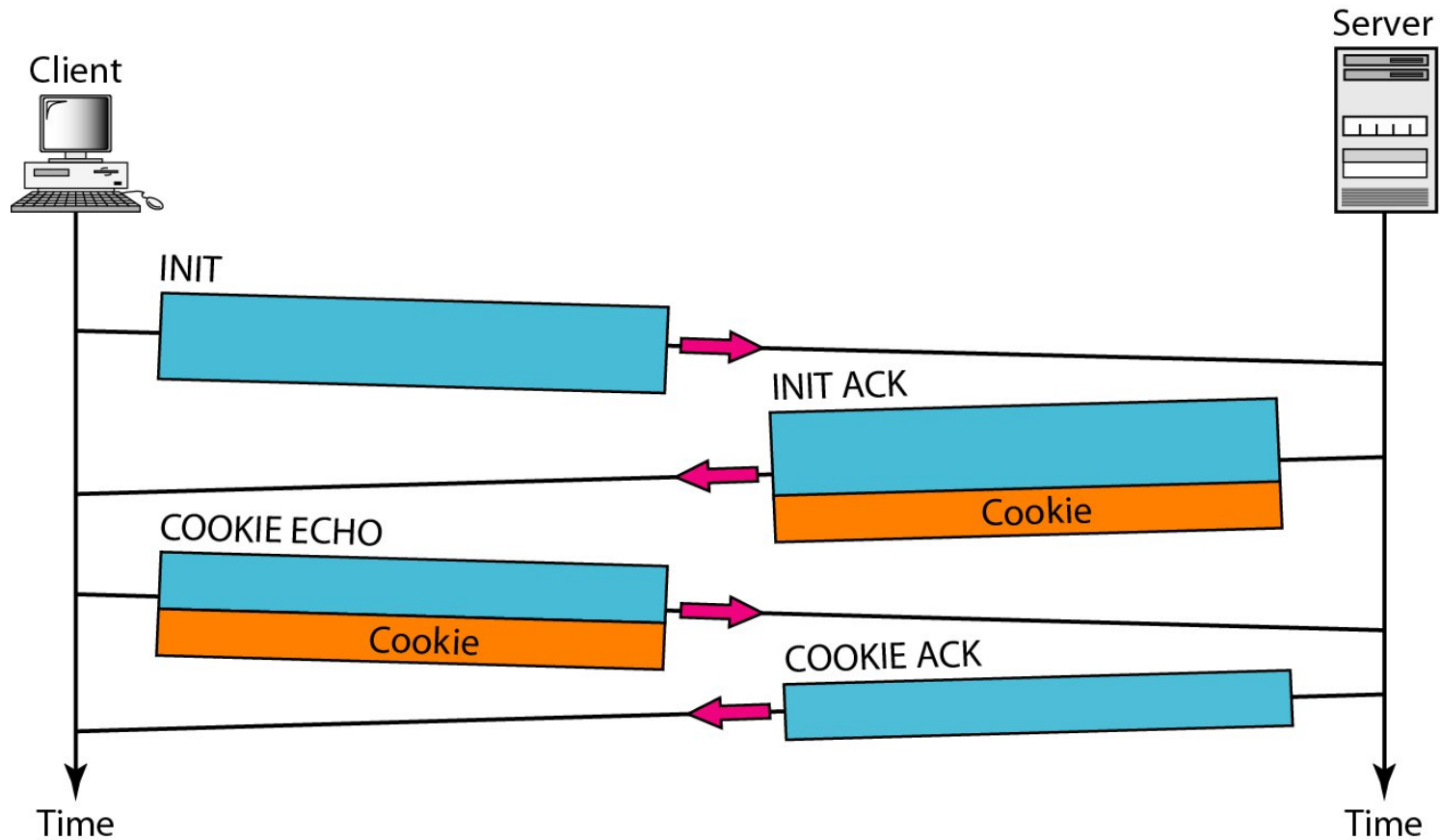


A connection in SCTP is called an association.

Note

No other chunk is allowed in a packet carrying an INIT or INIT ACK chunk. A COOKIE ECHO or a COOKIE ACK chunk can carry data chunks.

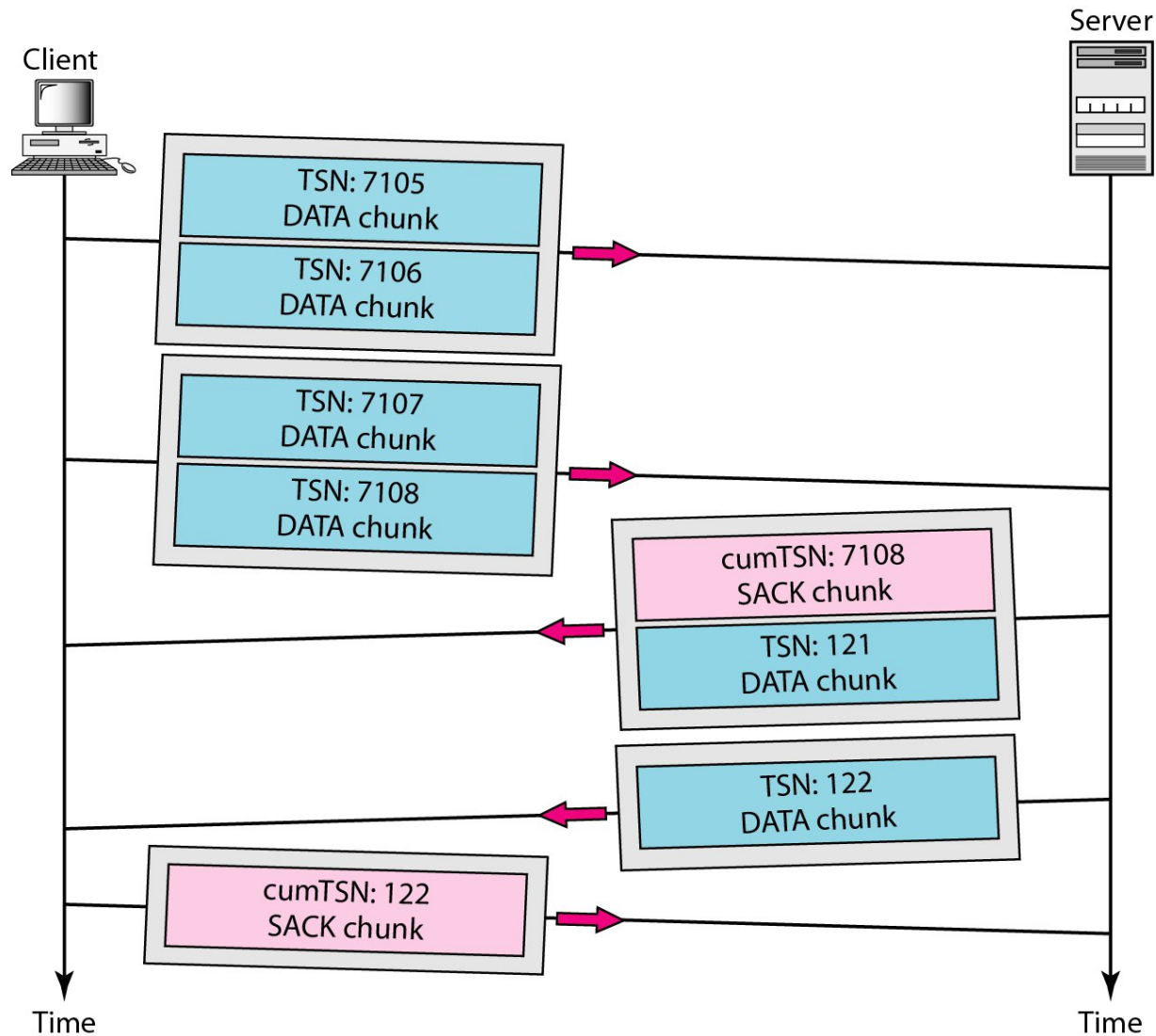
Figure 23.33 *Four-way handshaking*



Note

**In SCTP, only DATA chunks
consume TSNs;
DATA chunks are the only chunks
that are acknowledged.**

Figure 23.34 *Simple data transfer*



Note

The acknowledgment in SCTP defines the cumulative TSN, the TSN of the last data chunk received in order.

Figure 23.35 *Association termination*

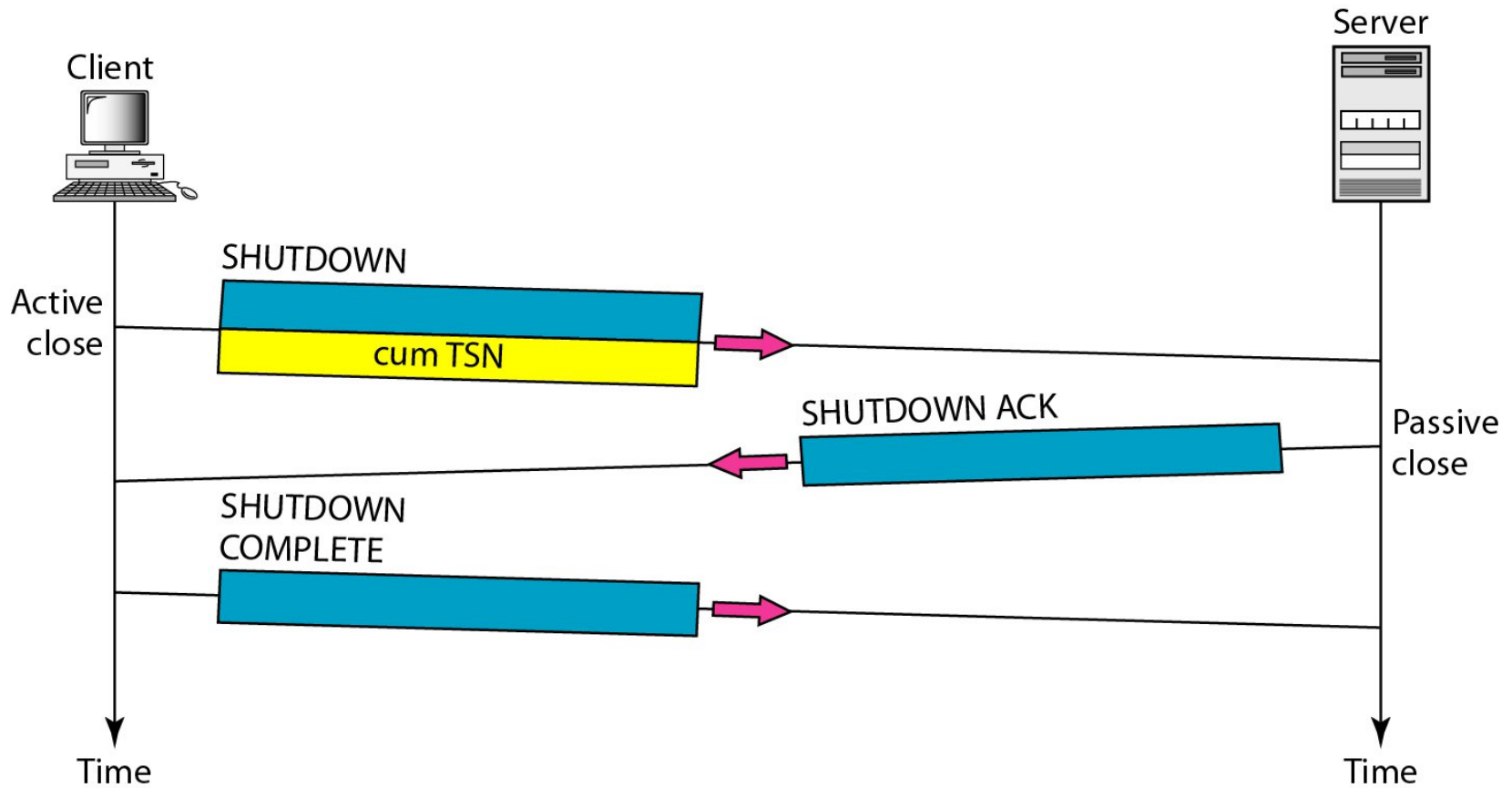


Figure 23.36 *Flow control, receiver site*

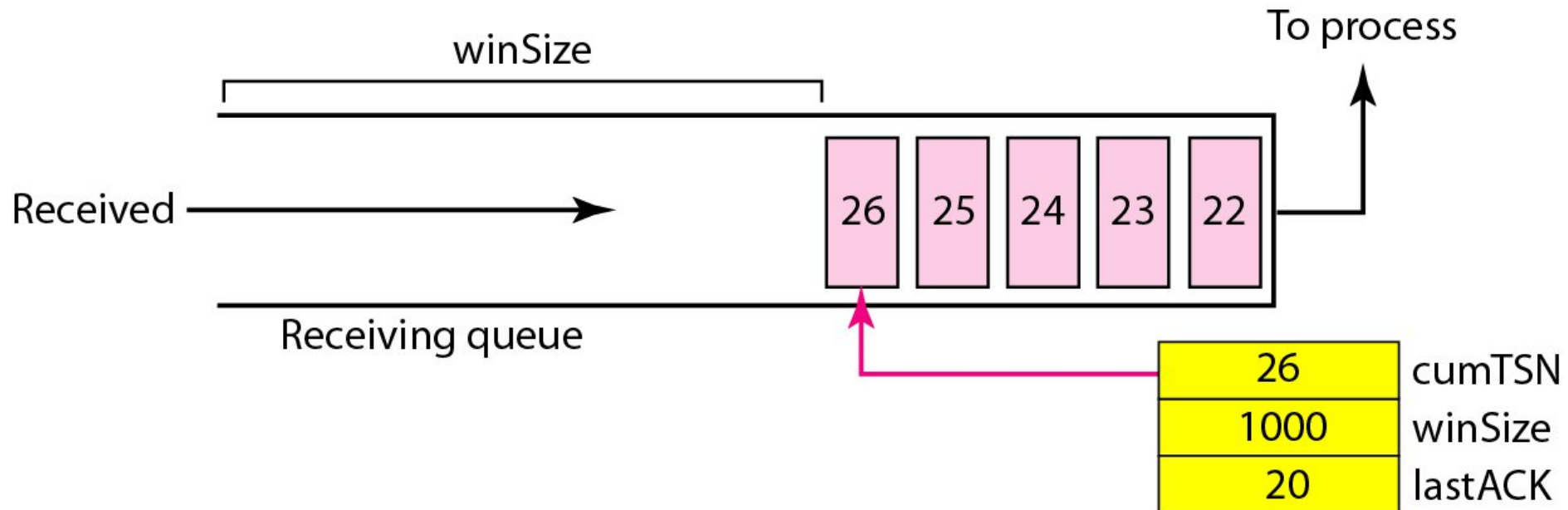


Figure 23.37 *Flow control, sender site*

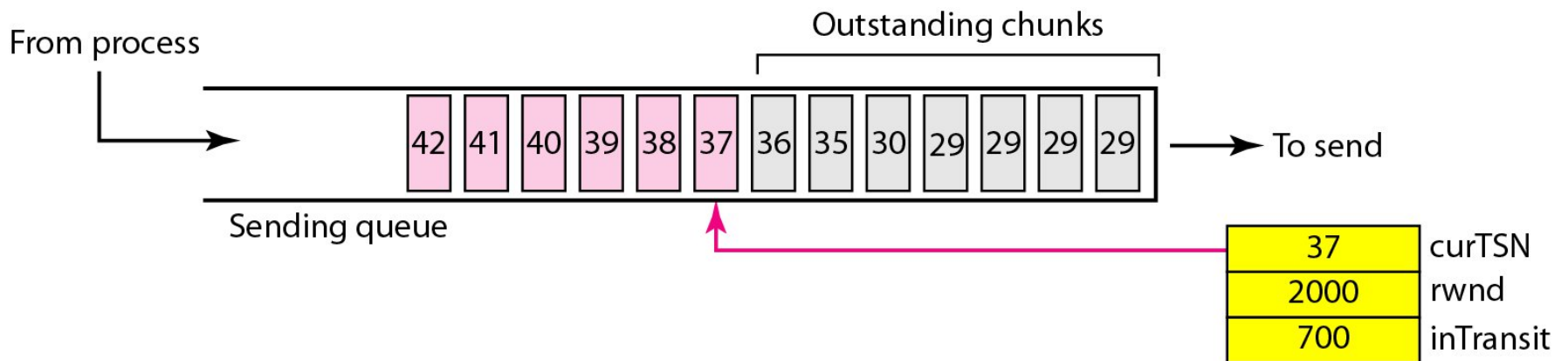


Figure 23.38 *Flow control scenario*

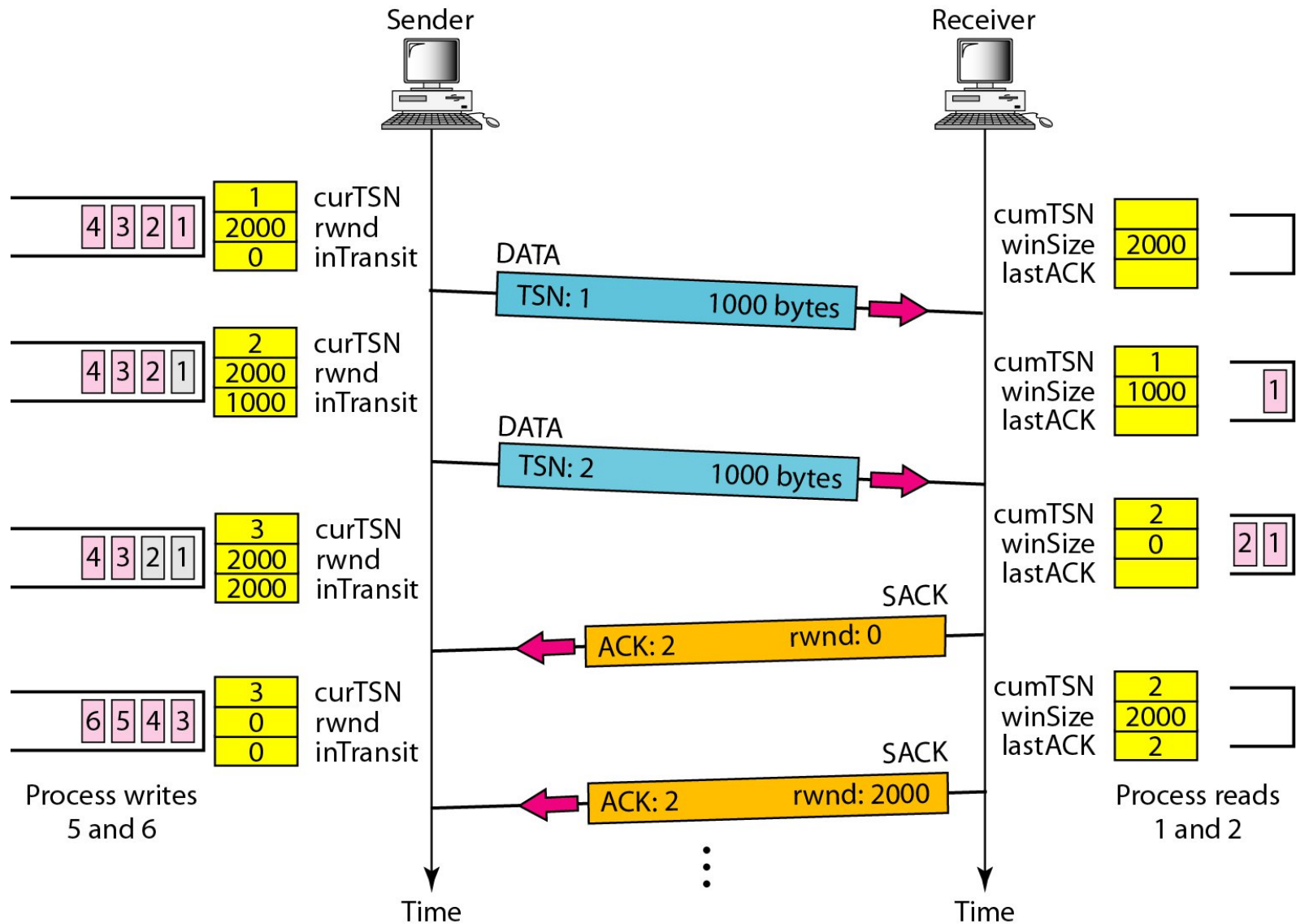


Figure 23.39 *Error control, receiver site*

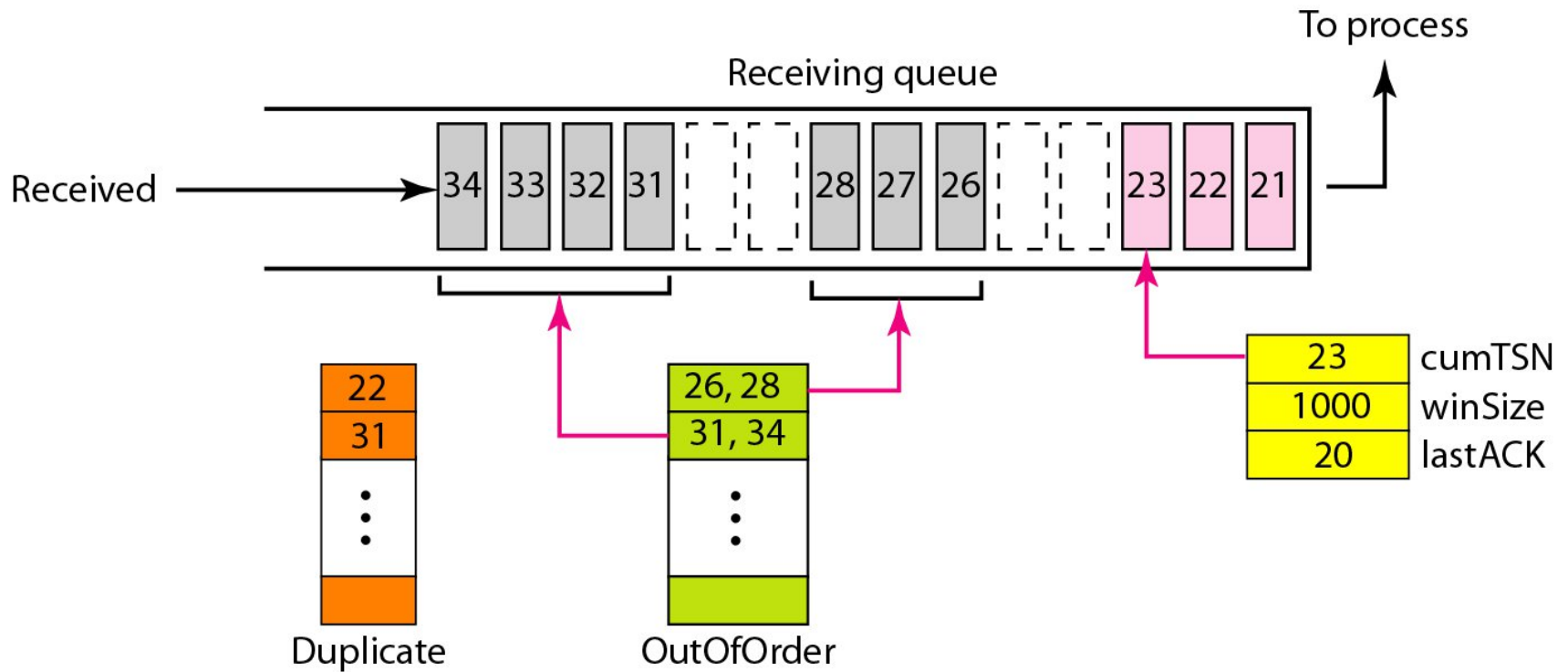


Figure 23.40 *Error control, sender site*

